

From Source to Execution: Design and Implementation of a Statically-Typed Programming Language with Type Inference

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ABSTRACT

This report presents the design and implementation of a small, statically-typed interpreted programming language realised through a four-stage compiler pipeline. The language provides four primitive types — Integer, Float, Boolean, and String — and supports arithmetic and equality expressions, variable assignment with type inference from first use, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in `print()` function. Types are inferred from program context rather than declared explicitly, combining static type safety with a declaration-free syntax. The pipeline consists of a hand-written lexer, a recursive-descent parser that constructs an abstract syntax tree, a static type checker that performs inference and rejects ill-typed programs before execution, and a tree-walking interpreter. Both a command-line runner and an interactive desktop interface expose each pipeline stage for inspection. The report documents the formal grammar, the type inference algorithm, the system architecture, and the automated test suite of 59 tests that validates the complete implementation.

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CHAPTER 1

INTRODUCTION

Programming languages define the formal contract between a programmer’s intent and machine execution. Building a language processor from source text through to evaluated output requires a coherent pipeline of components, each of which must correctly transform its input before passing it downstream. Lexical analysis identifies atomic tokens; parsing imposes hierarchical structure; static type checking validates semantic correctness before execution; and interpretation evaluates the validated program. When these stages are connected cleanly, errors are caught at the earliest possible point, and the overall system remains tractable to reason about, test, and extend.

This report presents the design and implementation of a small statically-typed interpreted language that realises this pipeline in full. The language supports four primitive types — Integer, Float, Boolean, and String — and provides arithmetic and equality expressions, assignment, conditional and iterative control flow, user-defined functions with value parameter passing, and a built-in `print()` statement. Types are inferred from program context rather than declared explicitly, combining the safety of static typing with the conciseness of a declaration-free syntax. The implementation spans three principal components developed in pipeline order: the lexer and symbol table, the recursive-descent parser and abstract syntax tree, and the type checker with interpreter. Each component was completed before the next was begun, since each stage consumes the structured output of its predecessor.

The implemented system can be invoked from the command line or through an interactive desktop interface, both of which surface the four pipeline stages — tokens, AST, inferred type table, and execution output — for inspection. An automated regression suite of 59 tests verifies correctness across all stages and language features.

The remainder of this report is organised as follows. Chapter 2 presents the language design, token specification, grammar, and operator precedence rules. Chapters 3 and 4 cover the front-end implementation in pipeline order: lexical analysis and the symbol table, then recursive-descent parsing and the AST. Chapter 5 explains the type inference algorithm and the semantic rules used by the checker. Chapter 6 presents the system architecture and implementation,

including the end-to-end pipeline, project layout, and technology stack. Chapter 7 discusses testing and verification, and Chapter 8 concludes the report.

CHAPTER 2

LANGUAGE DESIGN AND GRAMMAR

2.1 Types

The language provides four primitive types, chosen to cover the core domains of numeric computation, logical control flow, and text manipulation while keeping the type system tractable and the semantics well-defined.

2.1.1 *Integer*

The Integer type represents whole numbers such as 0, 7, and 42. Integer literals are written as one or more decimal digits without quotation marks or a decimal point.

2.1.2 *Float*

The Float type represents real-valued numbers written with a decimal point, such as 3.14 or 0.5. Float values are used exclusively with the dot-suffixed arithmetic operators (+., -., *., /.), following the Caml Light convention. Integer division (/) performs floor division and returns Integer: for example, 10 / 3 evaluates to 3 : Integer.

2.1.3 *Boolean*

The Boolean type represents logical truth values. The two Boolean literals are true and false. Boolean values are used in conditions and as results of comparison expressions.

2.1.4 *String*

The String type represents sequences of characters enclosed in double quotes, for example "hello". Strings can be assigned to variables and printed through the built-in print() function. The surrounding quotes are stripped by the parser when the Literal node is created, so the value stored internally is the bare character sequence without surrounding quotes. At runtime, the built-in print() function re-adds double quotes around string values, so print("plc") outputs "plc" : String.

2.1.5 Void

Although the language exposes exactly four programmer-visible types, the implementation introduces a fifth value, `Void`, within the `LanguageType` enumeration. Its sole purpose is to provide a well-defined return type for functions that contain no `return` statement, allowing the symbol table to record a consistent entry for every function regardless of whether it produces a value. `Void` cannot appear in source code and is never visible to the programmer; it is a bookkeeping device used internally by the type checker.

2.2 Static Typing

The language employs static typing, meaning that type errors are detected at type-checking time rather than at runtime. No explicit variable type declarations are required. Instead, the type checker infers types from literals, assignments, operations, function calls, and return statements. Once a variable type is established, later assignments must remain consistent with that type. For example, the following program is rejected at the type-checking stage before any execution occurs:

```
1 x = 5;
2 x = "hello";    (* TypeError: Cannot assign String to variable 'x' of
   type Integer *)
```

This contrasts with dynamically-typed languages, where the same program would run without error until (or unless) a type-dependent operation was attempted.

2.3 Token Specification

Table 2.1 summarises the major token classes used by the language.

Table 2.1. Token Specification

Token Category	Example Lexemes
INT_LITERAL	0, 10, 42
FLOAT_LITERAL	3.14, 20.5
BOOL_LITERAL	true, false
STRING_LITERAL	"hello"
IDENTIFIER	variable and function names such as x, add
Arithmetic operators (integer)	+, -, *, /
Float arithmetic operators	+, -, *, / .
Comparison operators	==, !=
Assignment operator	=
Keywords	if, else, while, def, return, print
Delimiters	() { } , ;

2.4 Context-Free Grammar

The language grammar can be described by the context-free grammar $G = (V_N, V_T, P, S)$, where V_N is the set of non-terminals, V_T is the set of terminals, P is the set of production rules, and S is the start symbol program. The grammar is implemented through recursive-descent parsing functions.

2.4.1 Program Structure

```

1 program  -> statement* EOF
2 statement -> assignment
3           | if_stmt
4           | while_stmt
5           | func_def
6           | print_stmt
7           | return_stmt
8           | expr ';'

```

2.4.2 Arithmetic Expressions

```

1 expr  -> term (( '+' | '-' | '+.' | '-.') term)*
2 term  -> factor (( '*' | '/' | '*.' | '/.') factor)*
3 factor -> '-' factor      (* integer unary negation *)
4       | '-.' factor      (* float unary negation *)
5       | INT_LITERAL
6       | FLOAT_LITERAL
7       | STRING_LITERAL
8       | BOOL_LITERAL
9       | IDENTIFIER
10      | IDENTIFIER '(' args? ')',
11      | '(' expr ')'

```

The operators `+`, `-`, `*`, and `/` operate exclusively on Integer operands. The dot-suffixed operators `+.` , `-.`, `*.`, and `/.` operate exclusively on Float operands. This follows the Caml Light convention of using distinct operator tokens for integer and float arithmetic, eliminating the need for implicit numeric promotion. Unary negation is similarly split: `-x` negates an Integer (desugared to `0 - x`), and `-.x` negates a Float (desugared to `0.0 -. x`).

2.4.3 Boolean Expressions

```

1 bool_expr -> expr '==' expr
2           | expr '!=' expr
3           | BOOL_LITERAL
4           | expr (* fallback: type checker enforces Boolean *)

```

Boolean expressions are only parsed as conditions inside `if` and `while` statements. Comparison results are not storable in variables directly; the grammar does not include `bool_expr` in the `factor` rule. This is consistent with the assignment specification, which describes Boolean expressions primarily as conditions.

If `bool_expr` parses an expression and finds no following `==` or `!=` operator, the expression is returned as-is and the type checker verifies that its inferred type is Boolean.

2.4.4 Statements

```

1 assignment -> IDENTIFIER '=' expr ';'
2 if_stmt    -> 'if' '(' bool_expr ')' block ('else' block)?
3 while_stmt -> 'while' '(' bool_expr ')' block
4 print_stmt -> 'print' '(' expr ')' ';'
5 return_stmt -> 'return' expr ';'
6 block      -> '{' statement* '}'

```

2.4.5 Function Definitions and Calls

```

1 func_def -> 'def' IDENTIFIER '(' params? ')' block
2 params   -> IDENTIFIER (',' IDENTIFIER)*
3 args     -> expr (',' expr)*

```

Function calls use value parameter passing. This means the argument values are evaluated before the call and copied into the function’s local environment.

2.5 Operator Precedence and Associativity

Operator precedence is encoded structurally in the recursive-descent parser. Comparison is lowest, addition and subtraction are next, and multiplication and division have the highest

precedence among binary operators. Arithmetic operators are left-associative.

Table 2.2. Operator Precedence (1 = lowest, 3 = highest) and Associativity

Precedence	Operators	Associativity
1 (lowest)	<code>==</code> , <code>!=</code>	non-associative (condition use only)
2	<code>+</code> , <code>-</code> , <code>+</code> , <code>.</code> , <code>-</code>	left-associative
3 (highest)	<code>*</code> , <code>/</code> , <code>*</code> , <code>.</code> , <code>/</code>	left-associative

CHAPTER 3

LEXICAL ANALYSIS AND SYMBOL TABLE

This chapter covers the responsibilities related to lexical and early semantic foundations of the language. It includes defining token structures and categories (such as keywords, operators, identifiers, and literals), implementing the lexer to transform source characters into tokens, and establishing the symbol table to manage variable and function storage along with their associated types. Together, these components ensure that raw source input is systematically organised into meaningful structures that can be used reliably by later stages of the compiler.

3.1 Token representation

The token system, written in `components/tokens.py`, establishes the vocabulary and structure of all lexical elements in the language. It specifies both the types of tokens that can exist and the structure each token must follow. This serves as a canonical contract between the lexer and the parser, ensuring consistency across all stages of the compiler.

Table 3.1 summarises the main categories, example lexemes, and typical `TokenType` values.

Table 3.1. Token categories and representative `TokenType` values

Category	Example lexemes	Representative <code>TokenType</code>
Literals	42, 3.14, true, "hello"	INT_LITERAL, FLOAT_LITERAL, BOOL_LITERAL, STRING_LITERAL
Identifiers	x, counter, add	IDENTIFIER
Keywords	if, else, while, def, return, print	keyword-specific token variants
Integer operators	+, -, *, /	PLUS, MINUS, STAR, SLASH
Float operators	+., -., *., /.	PLUS_DOT, MINUS_DOT, STAR_DOT, SLASH_DOT
Other operators	=, ==, !=	ASSIGN, EQUAL_EQUAL, BANG_EQUAL
Delimiters	Parentheses, braces, comma, and semicolon (see lexer delimiter tokens)	delimiter token variants
End marker	end of source input	EOF

Token types (from class `TokenType`) are defined using an Enum with `auto()` to automatically assign unique values. This avoids reliance on raw strings and ensures safer comparisons dur-

ing parsing. The token categories include literals, identifiers, keywords, operators, delimiters, and a special end-of-file (EOF) token. The EOF token is appended after all input is processed, allowing the parser to detect the end of input explicitly rather than encountering an unexpected termination.

Each token is represented as an immutable dataclass (`frozen=True`), meaning its fields cannot be modified after creation. Each token contains four key pieces of information: its kind (`token_type`), its lexeme (the exact substring from the source code), and its source position (`line`, `column`). This structured representation allows later stages to process the program in terms of syntax and errors without handling raw text directly. Immutability improves reliability by preventing accidental modification during later processing.

```
1 @dataclass(frozen=True)
2 class Token:
3     token_type: TokenType
4     lexeme: str
5     line: int
6     column: int
```

3.2 Lexer implementation

The lexer, implemented in `components/lexica.py`, is a hand-written scanner that converts raw source text into a sequence of tokens. Unlike generated lexers (for example SLY lexer mode, or Lex and Flex), this implementation uses explicit control flow (`if` and `while`) to process input.

The lexer maintains internal state including the source string, current position, start of the current token, line number, and column index. The column index is tracked per line and resets upon encountering a newline, while a separate `token_column` records the starting position of each token for accurate error reporting.

The main function, `tokenize`, iterates through the source code, repeatedly invoking `_scan_token` to identify the next token. Tokens are appended to a list, and an EOF token is added at the end.

Whitespace is skipped, while invalid input results in a `LexerError`.

```
1 def tokenize(self) -> list[Token]:
2     tokens: list[Token] = []
3     while not self._is_at_end():
4         self.start = self.current
5         self.token_column = self.column
6         token = self._scan_token()
7         if token is not None:
```

```

8         tokens.append(token)
9     tokens.append(Token(TokenType.EOF, "", self.line, self.column))
10    return tokens

```

Token recognition follows a structured approach. Single-character tokens are mapped using a lookup table. Multi-character operators such as `==` and `!=` are handled using lookahead via `_match`. String literals are processed by `_string`, which reads characters until a closing quotation mark or the end of input is reached, with error handling for unterminated strings. Numeric literals are handled by `_number`, which distinguishes integers and floating-point values based on the presence of a decimal point followed by digits. Identifiers are processed greedily, consuming alphanumeric characters and underscores, and then checked against the `KEYWORDS` map to determine whether they are reserved words or user-defined names. Notably, `true` and `false` are also entries in this map, mapping to `TokenType.BOOL_LITERAL` rather than a keyword-specific type. This means boolean literals are recognised through the same keyword-fallback mechanism as reserved words, preventing them from being scanned as user-defined identifiers.

```

1 KEYWORDS: dict[str, TokenType] = {
2     "if": TokenType.IF,
3     "else": TokenType.ELSE,
4     "while": TokenType.WHILE,
5     # ...
6     "true": TokenType.BOOL_LITERAL,
7     "false": TokenType.BOOL_LITERAL,
8 }

```

```

1 # Arithmetic operators with optional trailing '.' for float variant
2 _ARITH_DOT: dict[str, tuple[TokenType, TokenType]] = {
3     "+": (TokenType.PLUS, TokenType.PLUS_DOT),
4     "-": (TokenType.MINUS, TokenType.MINUS_DOT),
5     "*": (TokenType.STAR, TokenType.STAR_DOT),
6     "/": (TokenType.SLASH, TokenType.SLASH_DOT),
7 }

```

When one of these characters is encountered, the lexer checks the next character: if it is a `.` (and the character after that is not a digit), the dot-suffixed float variant is emitted; otherwise the plain integer operator is emitted. This lookahead prevents the `.` in a float literal such as `3.14` from being consumed as part of an operator.

```

1 SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
2     "=": TokenType.ASSIGN,
3     "(": TokenType.LPAREN,
4     ")": TokenType.RPAREN,
5     # ...
6 }

```


Helper functions such as `_advance`, `_peek`, and `_peek_next` manage character consumption and lookahead, while `_is_at_end` determines when input has been fully processed. Error messages include precise line and column information, aiding debugging and diagnostics.

Table 3.2 lists the main functions and data structures in `components/lexica.py` and their roles.

Table 3.2. Key functions and data in `lexica.py`

Function / data	Responsibility	Output / effect
<code>tokenize()</code>	Iterates through source, scans tokens, appends EOF	ordered token list
<code>_scan_token()</code>	Dispatches recognition based on character and lookahead	next token or skip
<code>_number()</code>	Parses numeric sequences, distinguishes int vs float	numeric literal token
<code>_string()</code>	Parses quoted text, checks termination	string token or error
<code>_identifier()</code>	Parses identifiers, resolves keyword vs user-defined	keyword or identifier token
<code>KEYWORDS</code> map	Maps reserved words to token types	token type selection
<code>SINGLE_CHAR_TOKENS</code> map	Maps single-character symbols	token type selection
<code>_advance()</code> , <code>_peek()</code> , <code>_match()</code>	Character consumption and lookahead	scanner state update

3.3 Symbol table and semantic structure

The symbol table, defined in `components/symbol_table.py`, operates at a higher level than the lexer. While the lexer processes raw text into tokens, the symbol table manages semantic information such as variable and function names, their types, and their scope.

The symbol table is implemented as a scoped structure, where each scope maintains its own dictionary of symbols and optionally references a parent scope. This enables proper handling of nested blocks and variable shadowing. Symbols are introduced through definition functions and retrieved through lookup operations, which search recursively through parent scopes if needed.

```

1 def lookup(self, name: str) -> Symbol:
2     symbol = self._symbols.get(name)
3     if symbol is not None:
4         return symbol
5     if self.parent is not None:
6         return self.parent.lookup(name)
7     raise SymbolTableError(f"Undefined symbol: {name}")

```

The type system is represented using a `LanguageType` enumeration, which includes fundamental types such as `Integer`, `Float`, `Boolean`, `String`, and `Void`. Symbols are modelled using an inheritance hierarchy: the base `Symbol` class stores a name and type, while subclasses such as `VariableSymbol` and `FunctionSymbol` include additional attributes. Variables track initialization state, while functions store parameter lists and return types. Table 3.3 summarises the main building blocks.

Table 3.3. Symbol table components and roles

Component	Stored information	Role in semantic analysis
Scoped symbol table	Symbols in current and parent scope	Supports nesting and shadowing via recursive lookup
<code>LanguageType</code> enum	<code>Integer</code> , <code>Float</code> , <code>Boolean</code> , <code>String</code> , <code>Void</code>	Defines the type system
<code>Symbol</code> (base)	Name and declared or inferred type	Base abstraction for entries
<code>VariableSymbol</code>	Variable metadata (including initialization state)	Tracks correct usage before assignment
<code>FunctionSymbol</code>	Parameters and return type	Validates calls and return consistency
<code>lookup(name)</code>	Recursive scope search	Resolves identifiers or reports undefined
<code>SymbolTableError</code>	Semantic error details	Reports duplicates, undefined symbols, invalid operations

Errors related to semantic violations are handled via `SymbolTableError`. These include duplicate declarations within the same scope, references to undefined symbols, and invalid operations such as attempting to mark a non-variable symbol as initialized.

3.4 Role of the symbol table in the pipeline

The symbol table is constructed during the type-checking phase rather than during lexical analysis. This guideline is followed in the implementation. The type checker initialises a global symbol table and populates it while traversing the AST, including defining variables and functions, managing nested scopes, and validating type correctness.

```

1 def run_pipeline(source_code: str) -> PipelineResult:
2     tokens = Lexer(source_code).tokenize()
3     tree = Parser(tokens).parse()
4     ast_text = ASTPrinter().print(tree)
5     checked_scope = TypeChecker().check(tree)
6     outputs: list[str] = []
7     Interpreter(output_callback=outputs.append).execute(tree)
8     return PipelineResult(
9         tokens=tokens,

```

```
10     ast_text=ast_text ,
11     checked_scope=checked_scope ,
12     outputs=outputs ,
13 )
```

Once type checking is complete, the populated symbol table is returned as part of the pipeline result (`checked_scope`). The checker creates child scopes for blocks and functions during analysis, but the displayed `checked_scope` table represents the global scope entries returned by the pipeline. It can then be displayed or used for further analysis.

This separation of concerns keeps lexical analysis focused on tokenization, while semantic meaning is enforced during type checking. For user-defined functions, definitions are first registered in the global scope with placeholder parameter and return types. These are inferred from the first valid call and refined through body checking; subsequent calls must match the inferred parameter types.

CHAPTER 4

RECURSIVE-DESCENT PARSING AND THE ABSTRACT SYNTAX TREE

This chapter presents the implementation of the recursive-descent parser and the associated abstract syntax tree (AST) model. Starting from lexer-produced tokens, the parser constructs a hierarchical Program AST that represents expressions, statements, and control-flow structure in a form suitable for downstream semantic analysis and execution. Implemented syntax coverage includes arithmetic expressions, assignments, if-then-else, while, function definitions, function calls, return, and built-in `print(...)` statements. Equality operators (`==`, `!=`) are parsed in condition contexts (if/while) through the parser's boolean-expression path.

4.1 Abstract syntax tree representation

The AST is defined as a collection of Python dataclasses representing the structural elements of the programming language. It serves as an intermediate representation between parsing and later stages such as type checking and interpretation.

A key distinction exists between tokens and AST nodes. Tokens are flat and describe individual lexical elements such as operators or identifiers. In contrast, the AST is hierarchical. For example, while a token may represent the symbol `+`, a `BinaryOp` node represents an entire addition operation, including its left and right operands as subtrees.

All AST nodes inherit from a common base class, allowing them to store positional metadata (line, column). This ensures that errors detected in later stages can still reference precise locations in the original source code.

```
1 @dataclass
2 class ASTNode:
3     """Base class for every AST node. Carries source location for
4         diagnostics."""
5     line: int = field(default=0, kw_only=True)
6     column: int = field(default=0, kw_only=True)
```

4.1.1 Expressions

Expressions represent computations or values within the program. A `Literal` node stores values known at parse time, including integers, floating-point numbers, booleans, and strings. String literals are stored without surrounding quotation marks, as these are removed during parsing.

```
1 if tok.token_type == TokenType.STRING_LITERAL:
2     self._advance()
3     return Literal(value=tok.lexeme[1:-1], line=tok.line, column=tok.
        column)
```

An `Identifier` node represents the use of a variable name within an expression, indicating evaluation rather than declaration. A `BinaryOp` node models arithmetic and comparison operations such as addition, subtraction, multiplication, division, and equality checks, storing the operator along with references to left and right operands. A `FunctionCall` node represents function invocation, consisting of the function name and a list of argument expressions.

```
1 @dataclass
2 class BinaryOp(ASTNode):
3     """An arithmetic or comparison binary operation.
4
5     op is one of: '+', '-', '*', '/', '+.', '-.', '*.', '/.', '==', '!=',
6     Example:  a + b    x == 0    a +. b
7     """
8     op: str
9     left: ASTNode
10    right: ASTNode
```

4.1.2 Statements

Statements represent actions or control flow within the program. An `Assign` node models variable assignment, pairing a variable name with a value expression, while `Print` and `Return` nodes each store a single expression and represent output and function return operations respectively. A `Block` node groups multiple statements within braces, representing a scoped execution sequence.

Control flow is handled via `If` and `While` nodes. The `If` node stores a condition, a then-block, and an optional else-block, while `While` stores a loop condition and body block. Function definitions are represented by `FunctionDef` nodes, which include the function name, a list of parameter names, and a body block. Parameter types are intentionally not assigned at parse time, as type inference is handled later by the type checker. Finally, `Program` serves as the

AST root, containing all top-level statements.

4.2 Parser implementation

The parser consumes the list of tokens generated by the lexer and produces a single Program AST node. It is implemented as a hand-written recursive-descent parser based on the project grammar. This avoids parser generators and instead relies on explicit functions for each grammar rule.

The parser maintains internal state consisting of the token list and a position index (`self._pos`) indicating the current token. Tokens are accessed via helper methods such as `_peek` and `_advance`. Parsing errors are reported using a custom `ParseError` exception with line and column information consistent with lexer diagnostics.

```
1 def parse(self) -> Program:
2     """Parse the full token stream and return the root Program node."""
3     line, col = self._peek().line, self._peek().column
4     statements: list[ASTNode] = []
5     while not self._is_at_end():
6         statements.append(self._statement())
7     return Program(statements=statements, line=line, column=col)
```

In addition to the main `parse()` entry point, the parser's statement dispatcher (`_statement`) implements a deterministic decision order over token patterns: function definition, conditional, loop, return, print, assignment, then expression statement fallback. This design is important because it prevents ambiguity between syntactically similar forms. In particular, assignments are recognized using one-token lookahead (`IDENTIFIER` followed by `=`), while identifier-led expressions that are not assignments are parsed as expression statements and must still end with a semicolon. This gives the parser full coverage of both declarative-style and expression-driven top-level statements without requiring a parser generator.

4.2.1 Statement parsing

The main parsing loop repeatedly processes statements until EOF, with the statement type determined via dispatch on the current token. Function definitions, conditionals, loops, returns, and prints each have dedicated parsing methods. Assignments are detected using one-token lookahead (`IDENTIFIER` followed by `ASSIGN`), preventing misclassification of function calls as assignments. If no specific statement pattern matches, the parser falls back to an expression statement and enforces a terminating semicolon.

```

1 # assignment: IDENTIFIER "=" expr ";"
2 if (
3     tok.token_type == TokenType.IDENTIFIER
4     and self._peek_next().token_type == TokenType.ASSIGN
5 ):
6     return self._assignment()
7 # fallback: bare expression statement expr ";"
8 node = self._expr()
9 self._expect(TokenType.SEMICOLON, "Expected ';' after expression")

```

4.2.2 Expression parsing and precedence

Expressions are parsed in layered precedence levels: `_expr` handles `+`, `-`, `+`, and `-`; `_term` handles `*`, `/`, `*`, and `/`; and `_factor` handles literals, identifiers and function calls, and parenthesized expressions. Boolean conditions for `if` and `while` are parsed by `_bool_expr`, which recognises `==` and `!=` after parsing the left-hand arithmetic expression. As a result, equality parsing is condition-focused in the current grammar implementation rather than a general infix operator available in every expression position. Function calls are recognised when an identifier is followed by `(`, with argument lists parsed as comma-separated expressions. Parenthesized expressions are supported to override precedence. Unary negation (`-x`, `-5`) is also handled at the `_factor` level: a leading `-` token causes the parser to recursively parse the operand and return `BinaryOp(0 - operand)`, so the existing type rules for subtraction cover negation without any additional inference code.

```

1 def _expr(self) -> ASTNode: # +, -
2     ...
3 def _term(self) -> ASTNode: # *, /
4     ...
5 def _factor(self) -> ASTNode: # unary -, literals, identifiers/calls,
6     grouped expr

```

Figure 4.1 shows the AST produced for the statement `z = x + y * 2;`. Because `_term` handles `*` before `_expr` handles `+`, the multiplication sub-tree is nested one level deeper, encoding the higher precedence of `*` over `+` without any explicit priority table at runtime.

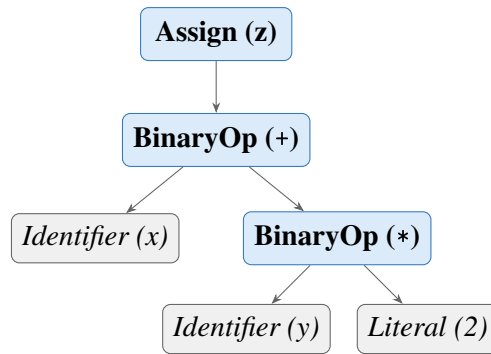


Figure 4.1. AST for $z = x + y * 2$; . Multiplication is a deeper sub-tree than addition, encoding higher precedence through tree structure rather than a runtime priority table.

Expression parsing is intentionally split into layered methods to enforce precedence and associativity in a transparent way. Parenthesized expressions are parsed explicitly and re-enter `_expr`, allowing precedence override where required by source syntax. Boolean conditions for `if` and `while` are parsed through `_bool_expr`, which accepts equality and inequality comparisons but can also return a non-comparison expression node; semantic enforcement that a condition is truly Boolean is deferred to the type checker.

4.2.3 Helper functions and lookahead

As described above, the parser uses `_check`, `_expect`, `_peek`, `_peek_next`, `_advance`, and `_is_at_end` for token navigation and validation. Lookahead is central for ambiguity resolution: assignment versus function call is resolved by inspecting the next token; comparison operators are recognised after parsing the left expression in `_bool_expr`. Parser errors follow a consistent diagnostic format: `[line X, col Y] ParseError: <message>`.

The parser also includes dedicated list parsers for formal parameters (`_params`) and call arguments (`_args`), both implemented as comma-separated sequences with optional emptiness. This ensures consistent handling of signatures such as `def f()` and `def f(a, b)` as well as invocations like `f()` and `f(x, y+1)`.

4.3 AST printing

For debugging and reporting, an AST printer converts the tree into a readable indented string, integrated into the pipeline output. The printer uses visitor-style dynamic dispatch, mapping node class names to methods such as `_visit_If` and `_visit_BinaryOp`, with output built as multi-line text where indentation reflects tree depth. Node formatting emphasises structure:

Program and Block include statement counts, BinaryOp explicitly shows left and right subtrees, and FunctionCall prints each argument by index. Literal includes both value and runtime Python type names for clarity.

4.4 Parser and type checker boundary

A deliberate architectural boundary is maintained between syntax construction and semantic validation. The parser is responsible for producing a structurally valid AST with accurate source locations, while type correctness and semantic constraints are checked later by the type checker. For example, `_bool_expr` may syntactically accept a non-Boolean expression as a condition node, but rejection of non-Boolean control-flow conditions is handled during semantic analysis. This separation keeps parser logic focused on grammar conformance and allows a clear responsibility split across team member contributions.

CHAPTER 5

TYPE INFERENCE ALGORITHM

5.1 Overview

Type inference in this project means that the programmer does not write explicit type declarations for variables or functions. Instead, the compiler infers types from assignments, literals, operations, function calls, and return statements. This keeps the language simple while still preserving static type safety.

The type checker runs after parsing. It receives the AST and validates the program before execution. Any inconsistent use of types is reported as a type error, and execution is stopped.

The interpreter stage does not repeat this validation: it walks the same AST with ordinary Python representations of values. Static type information remains in the symbol table produced by the checker; run-time values are not separately labelled with `LanguageType`, except that `print()` formats output by inspecting the value's Python type.

5.2 Type Environment

The type environment is represented by the symbol table. Conceptually, it is a mapping

$$\Gamma : \text{Identifier} \rightarrow \text{Type}$$

The environment grows as the checker visits the AST. On first assignment, a variable name is inserted with its inferred type. When a new block or a function call is entered, a child environment is created so that nested scopes can be checked independently while still accessing outer bindings when needed.

5.3 Type Inference Procedure

The type checker performs type inference by traversing the abstract syntax tree before interpretation. The algorithm is syntax-directed: each AST node is visited, the types of its child nodes are inferred first, and the current node is checked according to the language typing rules.

Algorithm TYPECHECK(*node*, Γ)

Input: *node* = current AST node; Γ = current type environment / symbol table

Output: inferred type of *node*, or `TypeError`

1. **If** *node* is a **Literal**: return the corresponding primitive type: `Integer`, `Float`, `Boolean`, or `String`.
2. **If** *node* is an **Identifier**: look up the identifier in Γ . If it is not found, report an undefined variable error. Otherwise, return its stored type.
3. **If** *node* is a **BinaryOp**:
 - Infer the type of the left operand and the right operand.
 - If the operator is `+`, `-`, `*`, or `/`: require both operands to be `Integer`; return `Integer`.
 - If the operator is `+`, `-`, `*`, or `/`: require both operands to be `Float`; return `Float`.
 - If the operator is `==` or `!=`: require both operands to be the same arithmetic type; return `Boolean`.
 - Otherwise, report a `TypeError`.
4. **If** *node* is an **Assignment**:
 - Infer the type of the right-hand-side expression.
 - If the variable is not yet in Γ : insert it with the inferred type.
 - Else: verify the new type matches the existing type; if not, report a `TypeError`.
5. **If** *node* is an **If** statement:
 - Infer the condition type; require it to be `Boolean`.
 - Type-check the then-block in a child environment.
 - If an else-block exists, type-check it in a child environment.
6. **If** *node* is a **While** statement:
 - Infer the condition type; require it to be `Boolean`.
 - Type-check the loop body in a child environment.
7. **If** *node* is a **FunctionDef**:
 - Register the function name in Γ .
 - Parameter types are initially unknown.
 - Return type is inferred from `return` statements in the body.

8. If *node* is a **FunctionCall**:

- Infer all argument types.
- If this is the first valid call: store the argument types as the function’s parameter types.
- Else: compare argument types with the stored parameter types.
- Type-check the function body using a local environment.
- Return the inferred function return type.

9. If *node* is a **Print** statement: infer the expression type; accept any of the four supported types.

5.4 Inference Rules

5.4.1 Literals

Literal nodes have direct types:

$$\Gamma \vdash 42 : \text{Integer}$$
$$\Gamma \vdash 3.14 : \text{Float}$$
$$\Gamma \vdash \text{true} : \text{Boolean}$$
$$\Gamma \vdash \text{“hi”} : \text{String}$$

5.4.2 No Operator Overloading

The language does not support operator overloading. Following the Caml Light convention, integer and float arithmetic use entirely separate operator tokens, so every operator has a single, fixed type signature:

- $+$, $-$, $*$, $/$ — require two `Integer` operands; result is `Integer`.
- $+. , -. , *. , /.$ — require two `Float` operands; result is `Float`.
- $+$ is not overloaded for string concatenation; applying it to a `String` is a type error.
- $==$ and $!=$ are not overloaded for `Boolean` or `String` equality.
- No implicit numeric promotion: `Integer + Float` is a type error.

Because every operator has a unique, unambiguous type derivable from its token alone, the compiler can always infer expression types without requiring explicit type declarations from the programmer.

5.4.3 Arithmetic Expressions

The integer operators `+`, `-`, `*`, and `/` require both operands to be `Integer` and always return `Integer` (including division, which uses integer floor division). The float operators `+`, `-`, `*`, and `/` require both operands to be `Float` and always return `Float`. Mixing an `Integer` and a `Float` with any operator is a type error.

Table 5.1. Arithmetic operator type rules

Left operand	Operator	Right operand	Result type
Integer	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code>	Integer	Integer
Float	<code>+</code> , <code>-</code> , <code>*</code> , <code>/</code>	Float	Float
Integer	any	Float	TypeError
Float	any	Integer	TypeError
Any	any	String or Boolean	TypeError

Unary negation

Unary integer negation (`-x`) is desugared by the parser to `0 - x`. Unary float negation (`- . x`) is desugared to `0 . 0 - . x`. The type checker therefore applies the standard binary rules in both cases.

5.4.4 Boolean Expressions

The comparison operators `==` and `!=` always produce `Boolean`. Both operators require two operands of the *same* arithmetic type — either both `Integer` or both `Float`; mixed-type comparisons such as `Integer == Float` are rejected. Comparisons involving `Boolean` or `String` operands are also rejected by the type checker. Although `Boolean` and `String` values exist in the language, equality and inequality are deliberately restricted to arithmetic expressions because the project specification defines basic `Boolean` expressions as equality and inequality between two arithmetic expressions.

5.4.5 Assignment Statement

Assignments define variable types on first use. If a variable does not yet exist in the type environment, the checker infers the type of the right-hand side and inserts the variable into the

environment. If the variable already exists, the new value must have the same type.

If $x \notin \Gamma$ and $\Gamma \vdash e : T$, then $\Gamma, x : T \vdash x = e$

If $x : T \in \Gamma$ and $\Gamma \vdash e : T$, then $\Gamma \vdash x = e$

5.4.6 *If-Then-Else*

The condition of an `if` statement must be `Boolean`. The then-block and else-block are checked in child scopes so that local operations can be validated cleanly.

$$\Gamma \vdash c : \text{Boolean} \quad \Gamma' \vdash B_{\text{then}} \quad \Gamma'' \vdash B_{\text{else}} \quad \Longrightarrow \quad \Gamma \vdash \text{if}(c) B_{\text{then}} \text{ else } B_{\text{else}}$$

5.4.7 *While-Loop*

The condition of a `while` loop must also be `Boolean`. The loop body is checked in a child scope and must not violate the types of variables already introduced in the surrounding environment.

$$\Gamma \vdash c : \text{Boolean} \quad \Gamma' \vdash B \quad \Longrightarrow \quad \Gamma \vdash \text{while}(c) B$$

5.4.8 *Function Definition and Call*

Function parameter types are inferred from the first call to the function. After that first call, subsequent calls must use arguments of the same types. The return type is inferred from the return statements in the function body. If the function returns inconsistent types, the checker reports an error.

For functions that are defined but never called, the type checker performs a post-pass after all top-level statements have been processed. It infers parameter types by scanning the body for arithmetic and comparison usage, then checks the full body with those inferred types. This ensures that type errors inside never-called function bodies are still detected before execution.

Recursive calls are intentionally outside the scope of this implementation; supporting them would require additional fixed-point inference or explicit recursion handling that is orthogonal to the core type-inference algorithm.

5.5 **Type Error Reporting**

Type errors are reported with source positions so that the user can locate the problem quickly. The format used by the system is:

[line X, col Y] TypeError: message

Examples include assigning a String to an Integer variable, using a non-Boolean condition in an if statement, or supplying the wrong argument type to a function.

5.6 Example Type Inference Traces

Consider the following program fragment:

```
1 x = 3;
2 y = 4;
3 z = x + y;
4 print(z);
```

The checker infers:

- x: Integer (from literal 3)
- y: Integer (from literal 4)
- x + y: Integer (Integer + Integer → Integer)
- z: Integer

For a float example using dot-suffixed operators:

```
1 def addf(a, b) {
2     return a +. b;
3 }
4
5 result = addf(2.0, 3.5);
```

The first function call infers a: Float and b: Float. The operator +. requires both operands to be Float, so the function return type is inferred as Float. The variable result is also inferred as Float.

CHAPTER 6

SYSTEM ARCHITECTURE AND IMPLEMENTATION

This chapter describes how the implementation is organised: the end-to-end processing flow, repository layout, technology choices, and the main components that realise static analysis and execution (type checker, interpreter, and user interfaces). Detailed treatment of lexical analysis, the symbol table, and parsing appears in Chapters 3 and 4; Chapter 5 presents the type-inference rules that the type checker implements.

6.1 Compiler pipeline

The system follows a four-stage pipeline. Lexical and syntactic processing are kept separate from semantic analysis and execution, which simplifies testing and localises errors to the appropriate stage.

The overall flow of data through the system is shown in Figure 6.1.

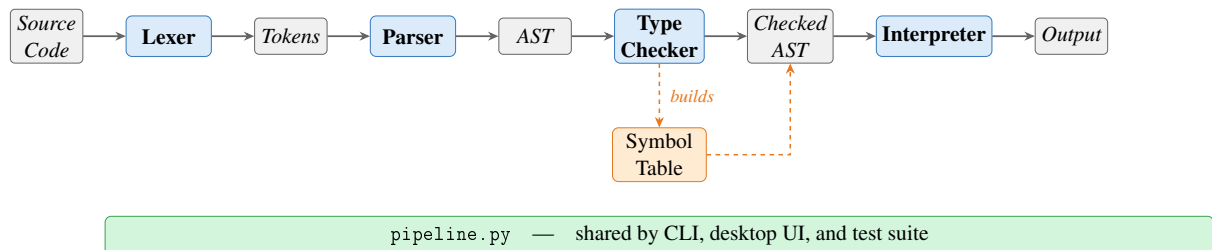


Figure 6.1. Compiler pipeline overview.

Source characters are scanned by the lexer into a token stream. The parser consumes tokens and builds an AST that reflects program structure (expressions, statements, and control flow). The type checker then walks the AST, populates and uses the symbol table, infers types, and validates scope and type rules. Only programs that pass the type checker proceed to interpretation. The interpreter evaluates the checked tree using ordinary Python values; it does not re-run the type rules, since correctness at runtime is guaranteed by the prior static pass. All four stages are orchestrated through `pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite.

6.1.1 End-to-end pipeline walkthrough

The following example traces the complete execution of a three-statement program through all four pipeline stages, illustrating the correspondence between source text, tokens, abstract syntax tree, inferred types, and runtime output.

Source program:

```
1 a = 3;
2 b = 4;
3 print(a + b);
```

Stage 1 — Lexer output (token stream):

1	IDENTIFIER	'a'	line 1, col 1
2	ASSIGN	'='	line 1, col 3
3	INT_LITERAL	'3'	line 1, col 5
4	SEMICOLON	';'	line 1, col 6
5	IDENTIFIER	'b'	line 2, col 1
6	ASSIGN	'='	line 2, col 3
7	INT_LITERAL	'4'	line 2, col 5
8	SEMICOLON	';'	line 2, col 6
9	PRINT	'print'	line 3, col 1
10	LPAREN	'('	line 3, col 6
11	IDENTIFIER	'a'	line 3, col 7
12	PLUS	'+'	line 3, col 9
13	IDENTIFIER	'b'	line 3, col 11
14	RPAREN	)'	line 3, col 12
15	SEMICOLON	';'	line 3, col 13
16	EOF		line 3, col 14

Stage 2 — Parser output (abstract syntax tree):

```
1 Program [3 statement(s)]
2   Assign 'a' =
3     Literal 3 (int)
4   Assign 'b' =
5     Literal 4 (int)
6   Print
7     BinaryOp '+'
8       left:
9         Identifier 'a'
10      right:
11        Identifier 'b'
```

Stage 3 — Type checker output (inferred type environment):

Name	Inferred type
a	Integer
b	Integer
a + b	Integer (Integer + Integer → Integer)

Stage 4 — Interpreter output:

```
1 7 : Integer
```

The example demonstrates the full data flow: source characters are tokenised by the lexer, structured into an AST by the parser, validated and annotated by the type checker, and finally evaluated by the interpreter to produce typed output.

6.1.2 Entry points and shared pipeline

All stages are orchestrated by `components/pipeline.py`, which is shared by the command-line runner, the desktop UI, and the test suite. From the project root, `python3 main.py` runs a script (or built-in sample) and prints the stages in order: tokens, AST text, type table, and execution output. `python3 ui.py` runs the Tkinter workbench: the same pipeline is used internally, with each stage shown in a separate tab so the user can inspect results without altering the underlying flow.

6.2 Project structure

The main source files and their responsibilities are summarised in Table 6.1.

Table 6.1. Main source files

File	Responsibility
<code>main.py</code>	Command-line runner for sample code or a script file
<code>ui.py</code>	Desktop UI for editing and running a script
<code>components/tokens.py</code>	Token classes and token categories
<code>components/lexica.py</code>	Lexical analysis
<code>components/ast_nodes.py</code>	AST node definitions
<code>components/parser.py</code>	Recursive-descent parser
<code>components/ast_printer.py</code>	AST renderer for debugging and reporting
<code>components/symbol_table.py</code>	Scoped symbol table and semantic types
<code>components/type_checker.py</code>	Static type checker and inference
<code>components/interpreter.py</code>	Tree-walking interpreter
<code>components/pipeline.py</code>	Shared pipeline for CLI, UI, and tests

6.3 Technology stack

The project uses Python and standard-library tools only (Table 6.2).

Table 6.2. Technology stack

Tool	Purpose
Python 3.10+	Core implementation language
<code>dataclasses</code>	AST and runtime helper structures
<code>tkinter</code>	Desktop graphical interface
<code>unittest</code>	Automated testing

6.4 Type checker

The `TypeChecker` class implements the inference and checking rules described in Chapter 5. It traverses the AST before execution, uses the symbol table to record variable and function information, infers types on first assignment, validates arithmetic and comparison expressions, enforces Boolean conditions for control flow, infers function parameter types from the first call (or from body usage for uncalled functions), and infers function return types from `return` statements. Programs that fail these rules are rejected before interpretation.

6.4.1 *Scope of the implemented type system*

In scope terms, this component turns a syntactically valid AST into a semantically valid program:

- static typing rules for arithmetic, comparison, assignment, control flow, and functions,
- type checking for expressions and statements,
- variable type inference from first assignment,
- function parameter inference from the first call (or from body usage for uncalled functions),
- function return type inference from `return` statements,
- source-positioned type errors when a program violates a typing rule.

6.5 Interpreter

The interpreter is a tree-walking evaluator over the checked AST. It maintains nested runtime environments for variables and function calls. Assignments write through the current or nearest enclosing scope; `if` selects a branch; `while` repeats until its condition becomes false; function calls bind arguments by value and return through an internal return mechanism.

The built-in `print()` function outputs both value and runtime type. All four types are supported:

```
1 print(42);           (* 42 : Integer      *)
2 print(3.14);         (* 3.14 : Float      *)
3 print(true);         (* true : Boolean    *)
4 print("plc");        (* "plc" : String   *)
```

6.5.1 Scope of the implemented runtime

The runtime stage provides observable behaviour and integrates with the shared pipeline:

- assignment execution,
- `if` execution,
- `while` execution,
- function execution with value parameter passing,
- `print()` execution with immediate output,
- execution behind the same `run_pipeline` path used by the CLI, UI, and tests.

6.6 Graphical user interface

The desktop workbench (`ui.py`) is implemented with Python's `tkinter` standard library and requires no additional dependencies. The window is titled *125970-126690-126687-Ass2-PLC* and opens at 1200×760 pixels in a horizontal split layout.

Toolbar. A top toolbar provides three buttons — *Open Script*, *Run*, and *Clear Output* — and a right-aligned status label that reports *Ready*, a loaded filename, or *Run failed*.

Left panel (source editor). A monospaced Text widget with both horizontal and vertical scrollbars allows the user to enter or paste arbitrary source code. The *Open Script* button

populates the editor from a file. The editor is pre-loaded with the built-in sample program on first launch.

Right panel (output tabs). A Notebook widget presents four tabs, each backed by a scrollable read-only Text widget with both horizontal and vertical scrollbars:

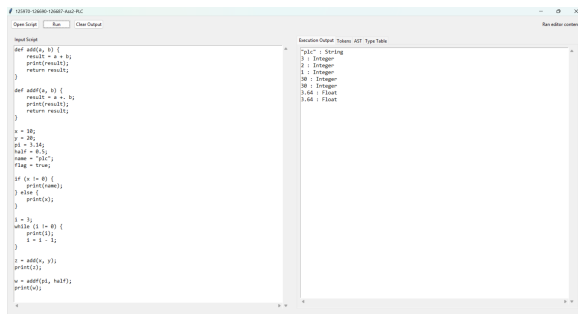
- *Execution Output* — the `print()` values produced by stage 4, formatted as `value : Type`;
- *Tokens* — the token stream from stage 1, one token per line with type, lexeme, and source position;
- *AST* — the indented tree text from stage 2;
- *Type Table* — the symbol table from stage 3 showing each name, kind, inferred type, initialisation state, and parameter list.

Error handling. If any pipeline stage raises an exception, the error message is written to the *Execution Output* tab as `ERROR: <message>`. No modal dialogs are used, so the user can edit and rerun without dismissing a popup.

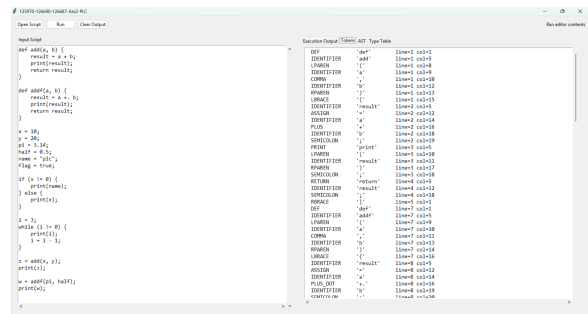
Keyboard shortcuts. F5 and Ctrl+Enter are both bound to the *Run* action, allowing rapid edit-run cycles without reaching for the mouse.

6.6.1 Screenshots

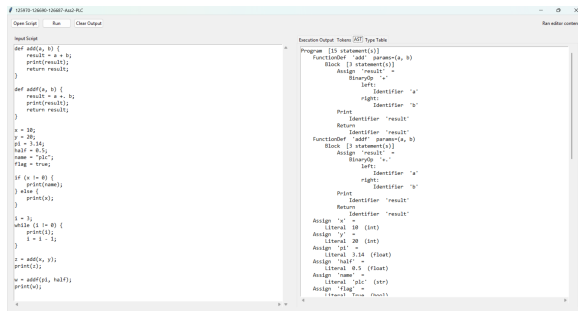
Figures 6.2 and 6.3 show the desktop workbench and CLI output.



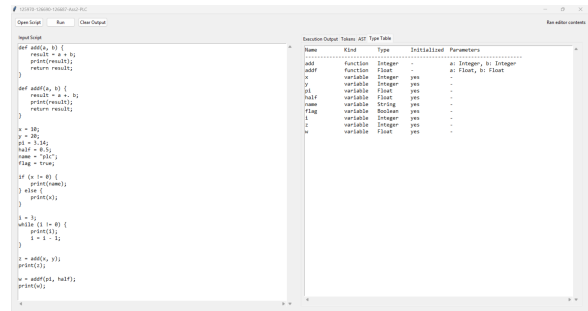
(a) Execution Output tab.



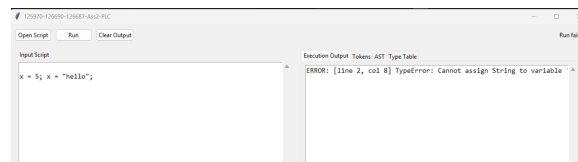
(b) Tokens tab (Stage 1).



(c) AST tab (Stage 2).



(d) Type Table tab (Stage 3).



(e) Inline type-error display.

Figure 6.2. Desktop workbench — output tabs and error display.

6.7 Command-line interface

The CLI runner (`main.py`) accepts an optional script path; without one it runs the built-in sample and prints four labelled stages to `stdout`.

```

PS C:\Users\winadm\Desktop\125970-126690-126687-Assignment2-PLC [set-externalenv] > .\ps C:\Users\winadm\Desktop\125970-126690-126687-Assignment2-PLC\venv\Scripts\Activate.ps1
(.venv) PS C:\Users\winadm\Desktop\125970-126690-126687-Assignment2-PLC> python main.py

=====
STAGE 1 - TOKENS
=====
def      'def'      line=2 col=1
IDENTIFIER 'add'      line=2 col=5
LPAREN   '('        line=2 col=8
IDENTIFIER 'a'       line=2 col=9
COMMA    ','        line=2 col=10
IDENTIFIER 'b'       line=2 col=12
RPAREN   ')'        line=2 col=13
LBRACE   '{'        line=2 col=15
IDENTIFIER 'result'  line=2 col=16
ASSIGN   '='        line=3 col=12
IDENTIFIER 'a'       line=3 col=14
PLUS     '+'        line=3 col=16
IDENTIFIER 'b'       line=3 col=18
SEMICOLON ';'       line=3 col=19
PRINT   'print'     line=4 col=5
LPAREN   '('        line=4 col=8
IDENTIFIER 'result'  line=4 col=11
RPAREN   ')'        line=4 col=13
SEMICOLON ';'       line=4 col=14
RETURN  'return'    line=5 col=5
IDENTIFIER 'result'  line=5 col=12
SEMICOLON ';'       line=5 col=14
Z       'z'         line=5 col=16
RBRACE   '}'        line=6 col=1
DEF      'def'      line=6 col=1
IDENTIFIER 'add'      line=6 col=5
LPAREN   '('        line=6 col=8
IDENTIFIER 'a'       line=6 col=9
COMMA    ','        line=6 col=10
IDENTIFIER 'b'       line=6 col=12
RPAREN   ')'        line=6 col=14
LBRACE   '{'        line=6 col=16
IDENTIFIER 'result'  line=6 col=17
ASSIGN   '='        line=6 col=18
IDENTIFIER 'a'       line=6 col=20
SEMICOLON ';'       line=6 col=21

```

(a) Stage 1: token stream.

```

Assign 'w' =
FunctionCall 'addf' [2 arg(s)]
arg[0]:
Identifier 'pi'
arg[1]:
Identifier 'half'

Print
Identifier 'w'

=====
STAGE 3 - TYPE CHECK
=====
Name      Kind      Type      Initialized Parameters
-----
add        function  Integer   -          a: Integer, b: Integer
addf       function  Float     -          a: Float, b: Float
x          variable  Integer   yes         -
y          variable  Integer   yes         -
pi         variable  Float     yes         -
half       variable  Float     yes         -
name       variable  String    yes         -
flag       variable  Boolean   yes         -
i          variable  Integer   yes         -
z          variable  Integer   yes         -
w          variable  Float     yes         -

=====
STAGE 4 - EXECUTION
=====
"plc" : String
3 : Integer
2 : Integer
1 : Integer
30 : Integer
30 : Integer
3.64 : Float
3.64 : Float
(.venv)[ps C:\Users\winadm\Desktop\125970-126690-126687-Assignment2-PLC>

```

(b) Stage 3 type table and Stage 4 output.

Figure 6.3. CLI run of python main.py on the built-in sample.

CHAPTER 7

TESTING AND VERIFICATION

7.1 Manual Testing

Prior to the construction of the automated suite, each language feature was exercised manually by running the full pipeline and inspecting stage outputs. Table 7.1 summarises the categories covered.

Table 7.1. Representative manually tested categories

Category	Program Behaviour	Expected Result
Arithmetic	Evaluate mixed numeric expressions	Correct value and inferred type
If-then-else	Choose one branch from a Boolean condition	Only matching branch runs
While-loop	Print during repeated execution	One output line per iteration
Functions	Infer parameters and return type	Valid call and correct return value
Type mismatch	Reassign incompatible type	Type error before runtime

7.1.1 *Lexer and Symbol Table*

The lexer was verified manually by running the full pipeline and inspecting Stage 1 token output for programs that exercise every token category: integer and float literals, Boolean literals, string literals, all six keywords, two-character comparison operators (`==`, `!=`), single-character operators, and delimiters. Identifier scanning with keyword fallback was confirmed by checking that `true`, `false`, and reserved words are emitted with their keyword token types while non-keyword names produce `IDENTIFIER` tokens. For example, the source fragment `x = 10;` produces the token sequence:

```

1 IDENTIFIER  'x'      line=1 col=1
2 ASSIGN      '='      line=1 col=3
3 INT_LITERAL '10'     line=1 col=5
4 SEMICOLON   ';'      line=1 col=7

```

Error paths were also checked: supplying a bare `!` character raises a `LexerError` with a source

position and a message suggesting !=, and an unterminated string literal raises a `LexerError` citing the opening line.

The symbol table was verified through the type checker and the Stage 3 output. Duplicate variable definitions were confirmed to raise a `SymbolTableError` before execution. Scope isolation was verified with nested blocks: a variable defined inside an `if` or `while` body is not visible in the outer scope after the block ends. The `format_table()` output was checked against the expected column layout (name, kind, type, initialisation status, parameters) for both variable and function symbols.

7.1.2 *Parser and AST*

The parser was verified manually by running the full pipeline and inspecting Stage 2 AST output. The following cases were checked:

- arithmetic expressions with mixed precedence (e.g. `x + y * 2`) produce a tree where multiplication is a deeper sub-tree than addition, confirming the `_expr/_term/_factor` precedence encoding,
- assignment versus expression statement dispatch: `x = 5`; produces an `Assign` node while `add(1,2)`; produces a `FunctionCall` node at the statement level, confirming the IDENTIFIER+ASSIGN lookahead,
- `if` with and without an `else` clause: the resulting `If` node has a non-null `else_block` only when the clause is present,
- function definitions store only parameter names in the `FunctionDef` node; the AST printer output was inspected to confirm no type annotations are present at this stage,
- string literals in `Literal` nodes have their surrounding double-quotes stripped; `print("hello")` prints `"hello"` : `String` at runtime (the interpreter re-adds the quotes when formatting string output),
- parse errors: missing semicolons, unmatched parentheses, and unexpected tokens all produce `[line X, col Y] ParseError:` messages with correct source positions.

7.1.3 Arithmetic Expressions

Arithmetic expressions were tested with integer-only and float-only programs to verify operator precedence and strict type separation. The expression `x + y * 2` confirmed that multiplication binds more tightly than addition, and mixing integer and float operands (e.g. `Integer + Float`) correctly produced a type error, confirming the no-overloading design.

7.1.4 Boolean Expressions

Boolean expressions were tested inside `if` and `while` conditions. The type checker correctly enforced that both operands of `==` and `!=` must be arithmetic; supplying a `Boolean` or `String` operand produced a source-positioned `TypeCheckError` before any execution.

7.1.5 Assignment Statements

Assignments were tested for both first-use type inference and reassignment consistency. Assigning `5` to a variable and subsequently assigning `"hello"` to the same variable correctly produced a type error, confirming that the inferred type is fixed after first assignment.

7.1.6 If-Then-Else

Conditional programs were constructed to verify that exactly one branch executes for a given condition and that neither branch is entered when the condition evaluates to the opposite truth value. Non-Boolean conditions were also supplied and confirmed to be rejected by the type checker before execution.

7.1.7 While-Loop

While-loops were tested with an integer countdown from three to zero. The results confirmed that the loop body executes once per iteration and terminates when the condition becomes false, with `print()` producing one output line per pass rather than accumulating output until loop exit.

7.1.8 Functions

Function tests covered value parameter passing, local scope execution, and return value propagation. The type checker correctly inferred parameter types from the first call site and enforced that signature for all subsequent calls to the same function.

7.1.9 *Print*

The built-in `print()` function was tested with all supported runtime types. The output includes the printed value together with its type, such as `3.14 : Float` and `"plc" : String`.

7.1.10 *Unary Minus*

The unary minus operator was verified for integer literals (`x = -5`), float literals (`y = -3.14`), negated variable references (`y = -x`), and negation inside a larger expression (`x = 3 + -2`). In each case the type checker assigned the correct type and the interpreter produced the expected value.

7.2 Type Error Detection

The checker was verified with invalid programs involving incompatible assignment, invalid arithmetic operands, wrong function arguments, and non-Boolean conditions. In each case the system stopped before execution and reported a source-positioned type error. Representative error messages are shown below.

```
1 # Assignment type mismatch
2 [line 2, col 1] TypeError: Cannot assign String to variable 'x' of type
   Integer
3
4 # Non-Boolean if condition
5 [line 1, col 4] TypeError: If condition must have type Boolean
6
7 # Wrong argument type for function
8 [line 5, col 9] TypeError: Argument 1 of 'add' must be Integer, got
   String
```

7.3 Automated Test Suite

The automated regression suite is located in `tests/test_pipeline.py` and uses Python's built-in `unittest` framework. The tests are organised into five classes, each targeting a distinct compiler stage.

Table 7.2. Automated test coverage by class

Test class	Stage covered	Tests
LexerTests	Tokenisation (<code>lexica.py</code>)	12
SymbolTableTests	Scoped symbol table (<code>symbol_table.py</code>)	7
ParserTests	Parsing and precedence (<code>parser.py</code>)	5
TypeCheckerTests	Static type inference (<code>type_checker.py</code>)	18
IntegrationTests	Full pipeline execution	17
Total		59

LexerTests verify that each token category (integer, float, Boolean, string, keywords, identifiers, two-character operators) is produced correctly, that source line numbers are tracked across newlines, and that illegal characters and unterminated strings raise `LexerError`.

SymbolTableTests exercise the symbol table directly, without going through the full pipeline. They cover variable definition and lookup, duplicate definition raising `SymbolTableError` for both variables and functions, parent-scope lookup through `child_scope()`, undefined-symbol lookup error, and the `format_table()` output for both variable and function symbol rows.

TypeCheckerTests verify static type inference and error detection. The 18 tests cover integer and float arithmetic (including dot-suffixed operators), integer floor division, strict type separation (mixed Integer/Float rejected for both arithmetic and comparison operators), string and boolean type inference, assignment type mismatch, non-Boolean conditions, undefined variables, wrong argument count and type, and two tests for uncalled-function body checking: one confirms that a type error inside a never-called function body is detected, and another confirms that a valid uncalled function raises no error.

IntegrationTests run programs through all four pipeline stages and verify the final output. Cases include unary negation on all numeric types, both branches of `if/else`, while-loop countdown, `print()` with every type, value-parameter isolation, nested function calls, variable reassignment, and scope isolation (variables declared inside an `if` block are not visible outside it).

The tests can be run with:

```
python3 -m unittest discover -s tests -v
```

All 59 tests pass.

7.4 Error Handling

The language system reports errors at three main stages:

- lexical errors for malformed characters or unterminated strings,
- parse errors for invalid syntax such as missing semicolons,
- type errors for programs that violate static typing rules.

Runtime errors are also reported with source positions whenever execution reaches an invalid state such as an undefined variable reference.

CHAPTER 8

CONCLUSION

This project delivered a complete, statically-typed interpreted language supporting Integer, Float, Boolean, and String types, with arithmetic, comparisons, assignments, if-then-else, while-loops, functions with value parameter passing, and a built-in `print()` function.

The system follows a four-stage pipeline: lexer, parser, type checker, and interpreter. Variable and function types are inferred without explicit declarations; programs that violate type rules are rejected before execution. For uncalled functions, the type checker performs a post-pass to infer parameter types from body usage, ensuring type errors are always caught.

The 59 automated tests across five classes — lexical analysis, symbol table, parsing, type checking, and full pipeline — confirm correct behaviour for all language features. Both a command-line runner and a desktop UI are provided.

The implementation demonstrates all core compiler-construction concepts covered in the course — lexical analysis, parsing, static type inference, and tree-walking interpretation — in a complete, tested, and working system. Future work could extend the language with recursive functions, richer data structures, and code generation.

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APPENDIX A: SOURCE CODE

The complete source code is available on GitHub:

<https://github.com/The-Token-Trio/125970-126690-126687-Assignment2-PLC>

The project can be run from the command line or the desktop UI:

```
1 python3 main.py
2 python3 ui.py
3 python3 -m unittest discover -s tests -v
```

No external Python dependencies are required.

APPENDIX B: TEAM MEMBERS AND CONTRIBUTIONS

Name	Student ID	Contributions
Aye Khin Khin Hpone (Yolanda Lim)	st125970	Static typing rules; type checking; assignment, if, while, function, and <code>print()</code> execution; unary minus inference; pipeline integration (<code>pipeline.py</code>); automated test suite (59 tests)
Applegate T. Tun Oo	st126690	Lexer implementation (scanning, tokenisation, line/column tracking, error reporting); token definitions; keywords, operators, identifiers, and literals; symbol table (variable/function/type storage)
Win Htut Naing	st126687	Arithmetic and boolean expressions; assignment, if-then-else, while-loop, function definition/call, and <code>print()</code> parsing; unary minus parsing

A.1 Entry Points

```
1 from __future__ import annotations
2
3 import argparse
4 from pathlib import Path
5
6 from components.pipeline import format_stage_output, run_pipeline
7
8
```



```

9  DEFAULT_SOURCE = """
10 def add(a, b) {
11     result = a + b;
12     print(result);
13     return result;
14 }
15
16 def addf(a, b) {
17     result = a +. b;
18     print(result);
19     return result;
20 }
21
22 x = 10;
23 y = 20;
24 pi = 3.14;
25 half = 0.5;
26 name = "plc";
27 flag = true;
28
29 if (x != 0) {
30     print(name);
31 } else {
32     print(x);
33 }
34
35 i = 3;
36 while (i != 0) {
37     print(i);
38     i = i - 1;
39 }
40
41 z = add(x, y);
42 print(z);
43
44 w = addf(pi, half);
45 print(w);
46 """
47
48
49 def run_source(source_code: str) -> None:
50     result = run_pipeline(source_code)
51     print(format_stage_output(result))
52
53
54 def _parse_args() -> argparse.Namespace:
55     parser = argparse.ArgumentParser(description="Run the programming
56         language pipeline.")
57     parser.add_argument("script", nargs="?", help="Optional path to a
58         source file to run")
59     return parser.parse_args()

```

```

60 def main() -> None:
61     args = _parse_args()
62     if args.script:
63         source_code = Path(args.script).read_text(encoding="utf-8")
64     else:
65         source_code = DEFAULT_SOURCE
66     run_source(source_code)
67
68
69 if __name__ == "__main__":
70     main()

```

Listing 8.1: main.py — command-line entry point

```

1  from __future__ import annotations
2
3  import tkinter as tk
4  from pathlib import Path
5  from tkinter import filedialog, ttk
6
7  from components.pipeline import format_tokens, run_pipeline
8
9
10 DEFAULT_SOURCE = """def add(a, b) {
11     result = a + b;
12     print(result);
13     return result;
14 }
15
16 def addf(a, b) {
17     result = a +. b;
18     print(result);
19     return result;
20 }
21
22 x = 10;
23 y = 20;
24 pi = 3.14;
25 half = 0.5;
26 name = \"plc\";
27 flag = true;
28
29 if (x != 0) {
30     print(name);
31 } else {
32     print(x);
33 }
34
35 i = 3;
36 while (i != 0) {
37     print(i);
38     i = i - 1;
39 }

```

```

40
41 z = add(x, y);
42 print(z);
43
44 w = addf(pi, half);
45 print(w);
46 """
47
48
49 class LanguageWorkbench:
50     def __init__(self) -> None:
51         self.root = tk.Tk()
52         self.root.title("125970-126690-126687-Ass2-PLC")
53         self.root.geometry("1200x760")
54         self.current_file: Path | None = None
55
56         self._build_layout()
57         self.source_text.insert("1.0", DEFAULT_SOURCE)
58         self.root.bind("<F5>", lambda _e: self._run_source())
59         self.root.bind("<Control-Return>", lambda _e: self._run_source
60             ())
61
62     def _build_layout(self) -> None:
63         toolbar = ttk.Frame(self.root, padding=10)
64         toolbar.pack(fill=tk.X)
65
66         ttk.Button(toolbar, text="Open Script", command=self.
67             _open_script).pack(side=tk.LEFT)
68         ttk.Button(toolbar, text="Run", command=self._run_source).pack(
69             side=tk.LEFT, padx=(8, 0))
70         ttk.Button(toolbar, text="Clear Output", command=self.
71             _clear_output).pack(side=tk.LEFT, padx=(8, 0))
72
73         self.status_var = tk.StringVar(value="Ready")
74         ttk.Label(toolbar, textvariable=self.status_var).pack(side=tk.
75             RIGHT)
76
77         body = ttk.Panedwindow(self.root, orient=tk.HORIZONTAL)
78         body.pack(fill=tk.BOTH, expand=True, padx=10, pady=(0, 10))
79
80         left_panel = ttk.Frame(body, padding=8)
81         right_panel = ttk.Frame(body, padding=8)
82         body.add(left_panel, weight=1)
83         body.add(right_panel, weight=1)
84
85         ttk.Label(left_panel, text="Input Script").pack(anchor=tk.W)
86         src_frame = ttk.Frame(left_panel)
87         src_frame.pack(fill=tk.BOTH, expand=True, pady=(6, 0))
88         self.source_text = tk.Text(src_frame, wrap=tk.NONE, font=(
89             "Consolas", 11))
90         src_vsb = ttk.Scrollbar(src_frame, orient=tk.VERTICAL, command=
91             self.source_text.yview)
92         src_hsb = ttk.Scrollbar(src_frame, orient=tk.HORIZONTAL,

```

```

        command=self.source_text.xview)
86 self.source_text.configure(yscrollcommand=src_vsb.set,
        xscrollcommand=src_hsb.set)
87 src_vsb.pack(side=tk.RIGHT, fill=tk.Y)
88 src_hsb.pack(side=tk.BOTTOM, fill=tk.X)
89 self.source_text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
90
91 notebook = ttk.Notebook(right_panel)
92 notebook.pack(fill=tk.BOTH, expand=True)
93
94 self.output_text = self._make_output_tab(notebook, "Execution
        Output")
95 self.tokens_text = self._make_output_tab(notebook, "Tokens")
96 self.ast_text = self._make_output_tab(notebook, "AST")
97 self.types_text = self._make_output_tab(notebook, "Type Table")
98
99 def _make_output_tab(self, notebook: ttk.Notebook, title: str) ->
    tk.Text:
100     frame = ttk.Frame(notebook, padding=8)
101     notebook.add(frame, text=title)
102     text = tk.Text(frame, wrap=tk.NONE, font=("Consolas", 11),
        state=tk.DISABLED)
103     vsb = ttk.Scrollbar(frame, orient=tk.VERTICAL, command=text.
        yview)
104     hsb = ttk.Scrollbar(frame, orient=tk.HORIZONTAL, command=text.
        xview)
105     text.configure(yscrollcommand=vsb.set, xscrollcommand=hsb.set)
106     vsb.pack(side=tk.RIGHT, fill=tk.Y)
107     hsb.pack(side=tk.BOTTOM, fill=tk.X)
108     text.pack(side=tk.LEFT, fill=tk.BOTH, expand=True)
109     return text
110
111 def _open_script(self) -> None:
112     file_path = filedialog.askopenfilename(
113         title="Open Script File",
114         filetypes=[("Text Files", "*.txt *.pl *.src"), ("All Files"
            , ".*")],
115     )
116     if not file_path:
117         return
118
119     path = Path(file_path)
120     self.current_file = path
121     self.source_text.delete("1.0", tk.END)
122     self.source_text.insert("1.0", path.read_text(encoding="utf-8")
        )
123     self.status_var.set(f"Loaded {path.name}")
124
125 def _run_source(self) -> None:
126     source = self.source_text.get("1.0", tk.END)
127     try:
128         result = run_pipeline(source)
129     except Exception as error:

```

```

130         self._write_text(self.output_text, f"ERROR: {error}")
131         self.status_var.set("Run failed")
132         return
133
134     self._write_text(self.output_text, "\n".join(result.outputs) if
135                     result.outputs else "(no output)")
136     self._write_text(self.tokens_text, format_tokens(result.tokens)
137                     )
138     self._write_text(self.ast_text, result.ast_text)
139     self._write_text(self.types_text, result.checked_scope.
140                     format_table())
141
142     current_label = self.current_file.name if self.current_file is
143                     not None else "editor contents"
144     self.status_var.set(f"Ran {current_label}")
145
146     def _clear_output(self) -> None:
147         self._write_text(self.output_text, "")
148         self._write_text(self.tokens_text, "")
149         self._write_text(self.ast_text, "")
150         self._write_text(self.types_text, "")
151         self.status_var.set("Output cleared")
152
153     @staticmethod
154     def _write_text(widget: tk.Text, content: str) -> None:
155         widget.config(state=tk.NORMAL)
156         widget.delete("1.0", tk.END)
157         widget.insert("1.0", content)
158         widget.config(state=tk.DISABLED)
159
160     def run(self) -> None:
161         self.root.mainloop()
162
163 if __name__ == "__main__":
164     LanguageWorkbench().run()

```

Listing 8.2: ui.py — Tkinter desktop workbench

A.2 Core Components

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass
4 from enum import Enum, auto
5
6
7 class TokenType(Enum):
8     # Literals
9     INT_LITERAL = auto()
10    FLOAT_LITERAL = auto()
11    BOOL_LITERAL = auto()

```

```

12  STRING_LITERAL = auto()
13
14  # Identifier
15  IDENTIFIER = auto()
16
17  # Keywords
18  IF = auto()
19  ELSE = auto()
20  WHILE = auto()
21  DEF = auto()
22  RETURN = auto()
23  PRINT = auto()
24
25  # Operators
26  PLUS = auto()
27  MINUS = auto()
28  STAR = auto()
29  SLASH = auto()
30  PLUS_DOT = auto()    # +. (float addition)
31  MINUS_DOT = auto()   # -. (float subtraction)
32  STAR_DOT = auto()    # *. (float multiplication)
33  SLASH_DOT = auto()   # /. (float division)
34  EQUAL_EQUAL = auto()
35  BANG_EQUAL = auto()
36  ASSIGN = auto()
37
38  # Delimiters
39  LPAREN = auto()
40  RPAREN = auto()
41  LBRACE = auto()
42  RBRACE = auto()
43  COMMA = auto()
44  SEMICOLON = auto()
45
46  EOF = auto()
47
48
49  @dataclass(frozen=True)
50  class Token:
51      token_type: TokenType
52      lexeme: str
53      line: int
54      column: int

```

Listing 8.3: components/tokens.py — token type enumeration and Token dataclass

```

1  from __future__ import annotations
2
3  from typing import Iterator
4
5  from components.tokens import Token, TokenType
6
7

```

```

8 KEYWORDS: dict[str, TokenType] = {
9     "if": TokenType.IF,
10    "else": TokenType.ELSE,
11    "while": TokenType.WHILE,
12    "def": TokenType.DEF,
13    "return": TokenType.RETURN,
14    "print": TokenType.PRINT,
15    "true": TokenType.BOOL_LITERAL,
16    "false": TokenType.BOOL_LITERAL,
17 }
18
19
20 # Arithmetic operators that may have a trailing '.' for the float
variant
21 _ARITH_DOT: dict[str, tuple[TokenType, TokenType]] = {
22     "+": (TokenType.PLUS, TokenType.PLUS_DOT),
23     "-": (TokenType.MINUS, TokenType.MINUS_DOT),
24     "*": (TokenType.STAR, TokenType.STAR_DOT),
25     "/": (TokenType.SLASH, TokenType.SLASH_DOT),
26 }
27
28 SINGLE_CHAR_TOKENS: dict[str, TokenType] = {
29     "=": TokenType.ASSIGN,
30     "(": TokenType.LPAREN,
31     ")": TokenType.RPAREN,
32     "{": TokenType.LBRACE,
33     "}": TokenType.RBRACE,
34     ",": TokenType.COMMA,
35     ";": TokenType.SEMICOLON,
36 }
37
38
39 class LexerError(Exception):
40     pass
41
42
43 class Lexer:
44     def __init__(self, source: str) -> None:
45         self.source = source
46         self.start = 0
47         self.current = 0
48         self.line = 1
49         self.column = 1
50         self.token_column = 1
51
52     def tokenize(self) -> list[Token]:
53         tokens: list[Token] = []
54         while not self._is_at_end():
55             self.start = self.current
56             self.token_column = self.column
57             token = self._scan_token()
58             if token is not None:
59                 tokens.append(token)

```

```

60     tokens.append(Token(TokenType.EOF, "", self.line, self.column))
61     return tokens
62
63     def _scan_token(self) -> Token | None:
64         char = self._advance()
65
66         # Whitespace handling
67         if char in {" ", "\t", "\r"}:
68             return None
69         if char == "\n":
70             self.line += 1
71             self.column = 1
72             return None
73
74         # Two-char operators
75         if char == "=":
76             if self._match("="):
77                 return self._make_token(TokenType.EQUAL_EQUAL)
78             return self._make_token(TokenType.ASSIGN)
79         if char == "!":
80             if self._match("="):
81                 return self._make_token(TokenType.BANG_EQUAL)
82             raise LexerError(self._error_message("Unexpected '!'. Did
83                 you mean '!='?"))
84
85         # Arithmetic operators: + - * / (with optional trailing '.'
86             for float variant)
87         arith = _ARITH_DOT.get(char)
88         if arith is not None:
89             int_tok, float_tok = arith
90             # Unary-minus lookahead: '-' followed by a digit is NOT
91                 '-.' even if
92             # next char is '.' -- we still need to check char after '.'
93                 is not digit
94             # (to avoid consuming the '.' of a float literal like 3.14)
95             .
96             # Rule: X. is a float operator only when the char after '.'
97                 is NOT a digit.
98             if self._peek() == "." and not self._peek_next().isdigit():
99                 self._advance() # consume '.'
100                 return self._make_token(float_tok)
101             return self._make_token(int_tok)
102
103         # Single-char operators/delimiters
104         single_char_token = SINGLE_CHAR_TOKENS.get(char)
105         if single_char_token is not None:
106             return self._make_token(single_char_token)
107
108         if char == '"':
109             return self._string()
110
111         if char.isdigit():
112             return self._number()

```



```

107
108     if self._is_identifier_start(char):
109         return self._identifier()
110
111     raise LexerError(self._error_message(f"Unexpected character: {
112         char!r}"))
113
114 def _string(self) -> Token:
115     while self._peek() != '"' and not self._is_at_end():
116         if self._peek() == "\n":
117             self.line += 1
118             self.column = 1
119             self._advance()
120
121     if self._is_at_end():
122         raise LexerError(self._error_message("Unterminated string
123             literal"))
124
125     # Closing quote
126     self._advance()
127     return self._make_token(TokenType.STRING_LITERAL)
128
129 def _number(self) -> Token:
130     while self._peek().isdigit():
131         self._advance()
132
133     is_float = False
134     if self._peek() == "." and self._peek_next().isdigit():
135         is_float = True
136         self._advance()
137         while self._peek().isdigit():
138             self._advance()
139
140     if is_float:
141         return self._make_token(TokenType.FLOAT_LITERAL)
142     return self._make_token(TokenType.INT_LITERAL)
143
144 def _identifier(self) -> Token:
145     while self._is_identifier_part(self._peek()):
146         self._advance()
147
148     lexeme = self._current_lexeme()
149     token_type = KEYWORDS.get(lexeme, TokenType.IDENTIFIER)
150     return self._make_token(token_type)
151
152 def _current_lexeme(self) -> str:
153     return self.source[self.start : self.current]
154
155 def _make_token(self, token_type: TokenType) -> Token:
156     return Token(token_type, self._current_lexeme(), self.line,
157         self.token_column)
158
159 def _advance(self) -> str:

```

```

157     char = self.source[self.current]
158     self.current += 1
159     self.column += 1
160     return char
161
162     def _match(self, expected: str) -> bool:
163         if self._is_at_end():
164             return False
165         if self.source[self.current] != expected:
166             return False
167
168         self.current += 1
169         self.column += 1
170         return True
171
172     def _peek(self) -> str:
173         if self._is_at_end():
174             return "\0"
175         return self.source[self.current]
176
177     def _peek_next(self) -> str:
178         if self.current + 1 >= len(self.source):
179             return "\0"
180         return self.source[self.current + 1]
181
182     def _is_at_end(self) -> bool:
183         return self.current >= len(self.source)
184
185     @staticmethod
186     def _is_identifier_start(char: str) -> bool:
187         return char.isalpha() or char == "_"
188
189     @staticmethod
190     def _is_identifier_part(char: str) -> bool:
191         return char.isalnum() or char == "_"
192
193     def _error_message(self, message: str) -> str:
194         return f"[line {self.line}, col {self.token_column}] {message}"
195
196
197     def iter_tokens(source: str) -> Iterator[Token]:
198         yield from Lexer(source).tokenize()

```

Listing 8.4: components/lexica.py — lexical analyser

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass, field
4 from typing import Optional
5
6
7 #

```

```

8  # Base
9  #
10
11 @dataclass
12 class ASTNode:
13     """Base class for every AST node. Carries source location for
14         diagnostics."""
15     line: int = field(default=0, kw_only=True)
16     column: int = field(default=0, kw_only=True)
17
18 #
19
20 # Expressions
21 #
22
23 @dataclass
24 class Literal(ASTNode):
25     """An integer, float, boolean, or string literal.
26
27     Examples:  42    3.14    true    "hello"
28
29     value: int | float | bool | str
30
31 @dataclass
32 class Identifier(ASTNode):
33     """A variable or function name used as an expression.
34
35     Example:  x
36
37     name: str
38
39 @dataclass
40 class BinaryOp(ASTNode):
41     """An arithmetic or comparison binary operation.
42
43     op is one of: '+' '-' '*' '/' '+.' '-.' '*.' '/.' '==' '!='
44     Example:  a + b    x == 0    a +. b
45
46     op: str
47     left: ASTNode
48     right: ASTNode
49
50 @dataclass
51
52

```

```

53 class FunctionCall(ASTNode):
54     """A call to a user-defined function.
55
56     Example:  add(1, 2)
57     """
58     name: str
59     args: list[ASTNode] = field(default_factory=list)
60
61
62 #
63 # -----
64 #
65 # Statements
66 # -----
67
68 @dataclass
69 class Assign(ASTNode):
70     """Variable assignment (first use infers type; later uses must
71     match).
72
73     Example:  x = 10;
74     """
75     name: str
76     value: ASTNode
77
78 @dataclass
79 class Print(ASTNode):
80     """Built-in print statement.
81
82     Example:  print(x);
83     """
84     expr: ASTNode
85
86 @dataclass
87 class Return(ASTNode):
88     """Return statement inside a function body.
89
90     Example:  return result;
91     """
92     expr: ASTNode
93
94 @dataclass
95 class Block(ASTNode):
96     """A brace-enclosed sequence of statements.
97
98     Example:  { x = 1; print(x); }
99     """
100     statements: list[ASTNode] = field(default_factory=list)

```

```

101
102
103 @dataclass
104 class If(ASTNode):
105     """If / optional else control flow.
106
107     Example:  if (x != 0) { print(x); } else { print(0); }
108     """
109     condition: ASTNode
110     then_block: Block
111     else_block: Optional[Block]
112
113
114 @dataclass
115 class While(ASTNode):
116     """While loop.
117
118     Example:  while (x != 0) { x = x - 1; }
119     """
120     condition: ASTNode
121     body: Block
122
123
124 @dataclass
125 class FunctionDef(ASTNode):
126     """Function definition.
127
128     Example:  def add(a, b) { return a + b; }
129
130     Note: parameter types are unknown at parse time -- Member 3's type
131           checker
132           will infer them from usage. We store only the parameter names here.
133     """
134     name: str
135     params: list[str]
136     body: Block
137
138 #
139 # -----
140 #
141
142 # Top-level
143 #
144 # -----
145
146 @dataclass
147 class Program(ASTNode):
148     """Root node: the entire source file as a list of statements."""
149     statements: list[ASTNode] = field(default_factory=list)

```

Listing 8.5: components/ast_nodes.py — AST node definitions

```

1 from __future__ import annotations
2
3 from typing import Optional
4
5 from components.tokens import Token, TokenType
6 from components.ast_nodes import (
7     ASTNode,
8     Program,
9     Block,
10    Assign,
11    If,
12    While,
13    FunctionDef,
14    FunctionCall,
15    Print,
16    Return,
17    BinaryOp,
18    Literal,
19    Identifier,
20 )
21
22
23 # -----
24 # Parser error
25 # -----
26
27 class ParseError(Exception):
28     pass
29
30
31 # -----
32 # Parser
33 # -----
34
35 class Parser:
36     """Recursive-descent parser.
37
38     Usage:
39         tokens = Lexer(source).tokenize()
40         tree   = Parser(tokens).parse()    # returns Program node
41     """
42
43     def __init__(self, tokens: list[Token]) -> None:
44         self._tokens = tokens

```

```

45     self._pos = 0
46
47     #
48     -----
49
50     # Public entry point
51     #
52     -----
53
54     def parse(self) -> Program:
55         """Parse the full token stream and return the root Program node
56         ."""
57         line, col = self._peek().line, self._peek().column
58         statements: list[ASTNode] = []
59         while not self._is_at_end():
60             statements.append(self._statement())
61         return Program(statements=statements, line=line, column=col)
62
63     #
64     -----
65
66     # Statements
67     #
68     -----
69
70     def _statement(self) -> ASTNode:
71         """Dispatch to the correct statement parser based on the
72         current token."""
73         tok = self._peek()
74
75         if tok.token_type == TokenType.DEF:
76             return self._func_def()
77
78         if tok.token_type == TokenType.IF:
79             return self._if_stmt()
80
81         if tok.token_type == TokenType.WHILE:
82             return self._while_stmt()
83
84         if tok.token_type == TokenType.RETURN:
85             return self._return_stmt()
86
87         if tok.token_type == TokenType.PRINT:
88             return self._print_stmt()
89
90         # assignment: IDENTIFIER "=" expr ";"
91         if (
92             tok.token_type == TokenType.IDENTIFIER
93             and self._peek_next().token_type == TokenType.ASSIGN
94         ):
95             return self._assignment()

```

```

88
89     # fallback: bare expression statement  expr ";"
90     node = self._expr()
91     self._expect(TokenType.SEMICOLON, "Expected ';' after
92         expression")
93     return node
94
95 def _assignment(self) -> Assign:
96     """IDENTIFIER '=' expr ';' """
97     name_tok = self._advance() #
98     IDENTIFIER
99     self._advance() # '='
100     value = self._expr()
101     self._expect(TokenType.SEMICOLON, "Expected ';' after
102         assignment")
103     return Assign(name=name_tok.lexeme, value=value,
104                   line=name_tok.line, column=name_tok.column)
105
106 def _if_stmt(self) -> If:
107     """'if' '(' bool_expr ')' block ( 'else' block )?"""
108     tok = self._advance() # 'if'
109     self._expect(TokenType.LPAREN, "Expected '(' after 'if'")
110     condition = self._bool_expr()
111     self._expect(TokenType.RPAREN, "Expected ')' after if condition
112         ")
113     then_block = self._block()
114     else_block: Optional[Block] = None
115     if self._check(TokenType.ELSE):
116         self._advance() # 'else'
117         else_block = self._block()
118     return If(condition=condition, then_block=then_block,
119              else_block=else_block,
120              line=tok.line, column=tok.column)
121
122 def _while_stmt(self) -> While:
123     """'while' '(' bool_expr ')' block"""
124     tok = self._advance() # '
125     while'
126     self._expect(TokenType.LPAREN, "Expected '(' after 'while'")
127     condition = self._bool_expr()
128     self._expect(TokenType.RPAREN, "Expected ')' after while
129         condition")
130     body = self._block()
131     return While(condition=condition, body=body,
132                  line=tok.line, column=tok.column)
133
134 def _func_def(self) -> FunctionDef:
135     """'def' IDENTIFIER '(' params? ')' block"""
136     tok = self._advance() # 'def'
137     name_tok = self._expect(TokenType.IDENTIFIER, "Expected
138         function name after 'def'")
139     self._expect(TokenType.LPAREN, "Expected '(' after function

```



```

        name")
132     params = self._params()
133     self._expect(TokenType.RPAREN, "Expected ')' after parameters")
134     body = self._block()
135     return FunctionDef(name=name_tok.lexeme, params=params, body=
        body,
136                           line=tok.line, column=tok.column)
137
138     def _print_stmt(self) -> Print:
139         """'print' '(' expr ')' ';'"""
140         tok = self._advance() # '
141         print'
142         self._expect(TokenType.LPAREN, "Expected '(' after 'print'")
143         expr = self._expr()
144         self._expect(TokenType.RPAREN, "Expected ')' after print
            expression")
145         self._expect(TokenType.SEMICOLON, "Expected ';' after print
            statement")
146         return Print(expr=expr, line=tok.line, column=tok.column)
147
148     def _return_stmt(self) -> Return:
149         """'return' expr ';'"""
150         tok = self._advance() # '
151         return'
152         expr = self._expr()
153         self._expect(TokenType.SEMICOLON, "Expected ';' after return
            value")
154         return Return(expr=expr, line=tok.line, column=tok.column)
155
156     def _block(self) -> Block:
157         """'{', statement* '}'"""
158         tok = self._expect(TokenType.LBRACE, "Expected '{'")
159         statements: list[ASTNode] = []
160         while not self._check(TokenType.RBRACE) and not self._is_at_end
            ():
161             statements.append(self._statement())
162             self._expect(TokenType.RBRACE, "Expected '}' to close block")
163         return Block(statements=statements, line=tok.line, column=tok.
            column)
164
165     #
166     -----
167
168     # Parameters (function definition)
169     #
170     -----
171
172     def _params(self) -> list[str]:
173         """IDENTIFIER (',' IDENTIFIER)* -- returns list of param names.
            """
174         params: list[str] = []
175         if self._check(TokenType.IDENTIFIER):

```

```

171         params.append(self._advance().lexeme)
172         while self._check(TokenType.COMMA):
173             self._advance()
174             tok = self._expect(TokenType.IDENTIFIER, "Expected
175                 parameter name after ','")
176             params.append(tok.lexeme)
177         return params
178
179 # -----
180
181 # Boolean expressions (lowest precedence, used in if/while
182 # conditions)
183 # -----
184
185 def _bool_expr(self) -> ASTNode:
186     """expr ('==' | '!=') expr | BOOL_LITERAL"""
187     # bare boolean literal: true / false
188     if self._check(TokenType.BOOL_LITERAL):
189         tok = self._advance()
190         value = tok.lexeme == "true"
191         return Literal(value=value, line=tok.line, column=tok.
192             column)
193
194     left = self._expr()
195
196     if self._check(TokenType.EQUAL_EQUAL):
197         op_tok = self._advance()
198         right = self._expr()
199         return BinaryOp(op="==", left=left, right=right,
200             line=op_tok.line, column=op_tok.column)
201
202     if self._check(TokenType.BANG_EQUAL):
203         op_tok = self._advance()
204         right = self._expr()
205         return BinaryOp(op="!=", left=left, right=right,
206             line=op_tok.line, column=op_tok.column)
207
208     # no comparison operator found -- return the expr as-is
209     # (the type checker will validate it's Boolean)
210     return left
211
212 # -----
213
214 # Arithmetic expressions (standard precedence)
215 # -----
216
217 def _expr(self) -> ASTNode:

```

```

213     """term (('+' | '-' | '+.' | '-.') term)*"""
214     node = self._term()
215     while self._check(TokenType.PLUS) or self._check(TokenType.
216         MINUS) \
217         or self._check(TokenType.PLUS_DOT) or self._check(
218             TokenType.MINUS_DOT):
219         op_tok = self._advance()
220         right = self._term()
221         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
222             line=op_tok.line, column=op_tok.column)
223     return node
224
225 def _term(self) -> ASTNode:
226     """factor (('*' | '/' | '*.' | '/.') factor)*"""
227     node = self._factor()
228     while self._check(TokenType.STAR) or self._check(TokenType.
229         SLASH) \
230         or self._check(TokenType.STAR_DOT) or self._check(
231             TokenType.SLASH_DOT):
232         op_tok = self._advance()
233         right = self._factor()
234         node = BinaryOp(op=op_tok.lexeme, left=node, right=right,
235             line=op_tok.line, column=op_tok.column)
236     return node
237
238 def _factor(self) -> ASTNode:
239     """
240     '-' factor (unary negation, desugared to 0 - factor)
241     | INT_LITERAL | FLOAT_LITERAL | STRING_LITERAL | BOOL_LITERAL
242     | IDENTIFIER ( '(' args? ') ' )?
243     | '(' expr ')'
244     """
245     tok = self._peek()
246
247     # Unary minus '-' factor (integer negation, desugared to 0 -
248     # factor)
249     if tok.token_type == TokenType.MINUS:
250         op_tok = self._advance()
251         operand = self._factor()
252         zero = Literal(value=0, line=op_tok.line, column=op_tok.
253             column)
254         return BinaryOp(op="-", left=zero, right=operand,
255             line=op_tok.line, column=op_tok.column)
256
257     # Unary minus-dot '-.' factor (float negation, desugared to
258     # 0.0 -. factor)
259     if tok.token_type == TokenType.MINUS_DOT:
260         op_tok = self._advance()
261         operand = self._factor()
262         zero_f = Literal(value=0.0, line=op_tok.line, column=op_tok.
263             column)
264         return BinaryOp(op="-.", left=zero_f, right=operand,
265             line=op_tok.line, column=op_tok.column)

```

```

258
259 # Integer literal
260 if tok.token_type == TokenType.INT_LITERAL:
261     self._advance()
262     return Literal(value=int(tok.lexeme), line=tok.line, column
                    =tok.column)
263
264 # Float literal
265 if tok.token_type == TokenType.FLOAT_LITERAL:
266     self._advance()
267     return Literal(value=float(tok.lexeme), line=tok.line,
                    column=tok.column)
268
269 # Boolean literal
270 if tok.token_type == TokenType.BOOL_LITERAL:
271     self._advance()
272     return Literal(value=(tok.lexeme == "true"), line=tok.line,
                    column=tok.column)
273
274 # String literal (keep the surrounding quotes stripped)
275 if tok.token_type == TokenType.STRING_LITERAL:
276     self._advance()
277     return Literal(value=tok.lexeme[1:-1], line=tok.line,
                    column=tok.column)
278
279 # Identifier -- could be a plain variable or a function call
280 if tok.token_type == TokenType.IDENTIFIER:
281     self._advance()
282     if self._check(TokenType.LPAREN):
283         # function call
284         self._advance() # '('
285         args = self._args()
286         self._expect(TokenType.RPAREN, "Expected ')' after
                        function arguments")
287         return FunctionCall(name=tok.lexeme, args=args,
                             line=tok.line, column=tok.column)
288     return Identifier(name=tok.lexeme, line=tok.line, column=
                    tok.column)
289
290 # Grouped expression '(' expr ')'
291 if tok.token_type == TokenType.LPAREN:
292     self._advance() # '('
293     node = self._expr()
294     self._expect(TokenType.RPAREN, "Expected ')' to close
                        grouped expression")
295     return node
296
297
298 raise ParseError(self._error_at(tok, f"Unexpected token: {tok.
    lexeme!r}"))
299
300 #

```

```

301 # Arguments (function call)
302 #
303 -----
304 def _args(self) -> list[ASTNode]:
305     """expr (',' expr)"""
306     args: list[ASTNode] = []
307     if not self._check(TokenType.RPAREN):
308         args.append(self._expr())
309         while self._check(TokenType.COMMA):
310             self._advance() # ','
311             args.append(self._expr())
312     return args
313
314 #
315 -----
316 # Token navigation helpers
317 #
318 -----
319
320 def _peek(self) -> Token:
321     return self._tokens[self._pos]
322
323 def _peek_next(self) -> Token:
324     if self._pos + 1 < len(self._tokens):
325         return self._tokens[self._pos + 1]
326     return self._tokens[-1] # EOF
327
328 def _advance(self) -> Token:
329     tok = self._tokens[self._pos]
330     if not self._is_at_end():
331         self._pos += 1
332     return tok
333
334 def _check(self, token_type: TokenType) -> bool:
335     return self._peek().token_type == token_type
336
337 def _is_at_end(self) -> bool:
338     return self._peek().token_type == TokenType.EOF
339
340 def _expect(self, token_type: TokenType, message: str) -> Token:
341     if self._check(token_type):
342         return self._advance()
343     raise ParseError(self._error_at(self._peek(), message))
344
345 def _error_at(self, tok: Token, message: str) -> str:
346     return f"[line {tok.line}, col {tok.column}] ParseError: {message}"

```

Listing 8.6: components/parser.py — recursive-descent parser

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass, field
4 from enum import Enum
5
6
7 class LanguageType(Enum):
8     INTEGER = "Integer"
9     FLOAT = "Float"
10    BOOLEAN = "Boolean"
11    STRING = "String"
12    VOID = "Void"
13
14
15 @dataclass
16 class Symbol:
17     name: str
18     symbol_type: LanguageType
19
20
21 @dataclass
22 class VariableSymbol(Symbol):
23     initialized: bool = False
24
25
26 @dataclass
27 class FunctionSymbol(Symbol):
28     parameters: list[tuple[str, LanguageType]] = field(default_factory=
29         list)
30
31 class SymbolTableError(Exception):
32     pass
33
34
35 class SymbolTable:
36     def __init__(self, parent: SymbolTable | None = None) -> None:
37         self.parent = parent
38         self._symbols: dict[str, Symbol] = {}
39
40     def define_variable(self, name: str, symbol_type: LanguageType,
41         initialized: bool = False) -> VariableSymbol:
42         if name in self._symbols:
43             raise SymbolTableError(f"Symbol already defined in this
44                 scope: {name}")
45         symbol = VariableSymbol(name=name, symbol_type=symbol_type,
46             initialized=initialized)
47         self._symbols[name] = symbol
48         return symbol
49
50     def define_function(
51         self,

```

```

49     name: str,
50     return_type: LanguageType,
51     parameters: list[tuple[str, LanguageType]],
52 ) -> FunctionSymbol:
53     if name in self._symbols:
54         raise SymbolTableError(f"Symbol already defined in this
55                                 scope: {name}")
56     symbol = FunctionSymbol(name=name, symbol_type=return_type,
57                             parameters=parameters)
58     self._symbols[name] = symbol
59     return symbol
60
61 def lookup(self, name: str) -> Symbol:
62     symbol = self._symbols.get(name)
63     if symbol is not None:
64         return symbol
65     if self.parent is not None:
66         return self.parent.lookup(name)
67     raise SymbolTableError(f"Undefined symbol: {name}")
68
69 def exists_in_current_scope(self, name: str) -> bool:
70     return name in self._symbols
71
72 def child_scope(self) -> SymbolTable:
73     return SymbolTable(parent=self)
74
75 def set_initialized(self, name: str) -> None:
76     symbol = self.lookup(name)
77     if not isinstance(symbol, VariableSymbol):
78         raise SymbolTableError(f"Cannot initialize non-variable
79                                 symbol: {name}")
80     symbol.initialized = True
81
82 def snapshot(self) -> dict[str, Symbol]:
83     return dict(self._symbols)
84
85 def format_table(self) -> str:
86     lines = [
87         f"{'Name':<12} {'Kind':<10} {'Type':<10} {'Initialized'
88         ':<12} Parameters",
89         "-" * 72,
90     ]
91
92     for name, symbol in self._symbols.items():
93         if isinstance(symbol, FunctionSymbol):
94             params = ", ".join(f"{param_name}: {param_type.value}"
95                                for param_name, param_type in symbol.parameters)
96             lines.append(
97                 f"{name:<12} {'function':<10} {symbol.symbol_type.
98                 value:<10} {'-':<12} {params or '-'}"
99             )
100         elif isinstance(symbol, VariableSymbol):
101             initialized = "yes" if symbol.initialized else "no"

```

```

96         lines.append(
97             f"{name:<12} {'variable':<10} {symbol.symbol_type.
              value:<10} {initialized:<12} -"
98         )
99     else:
100         lines.append(f"{name:<12} {'symbol':<10} {symbol.
                      symbol_type.value:<10} {'-':<12} -")
101
102     return "\n".join(lines)

```

Listing 8.7: components/symbol_table.py — scoped symbol table

```

1  from __future__ import annotations
2
3  from dataclasses import dataclass
4
5  from components.ast_nodes import (
6      ASTNode,
7      Assign,
8      BinaryOp,
9      Block,
10     FunctionCall,
11     FunctionDef,
12     Identifier,
13     If,
14     Literal,
15     Print,
16     Program,
17     Return,
18     While,
19 )
20 from components.symbol_table import FunctionSymbol, LanguageType,
    SymbolTable, SymbolTableError, VariableSymbol
21
22
23 class TypeCheckError(Exception):
24     pass
25
26
27 @dataclass
28 class _FunctionState:
29     node: FunctionDef
30     body_checked: bool = False
31
32
33 @dataclass
34 class _FunctionContext:
35     name: str
36     return_type: LanguageType | None = None
37
38
39 class TypeChecker:
40     def __init__(self) -> None:

```



```

41     self.global_scope = SymbolTable()
42     self._functions: dict[str, _FunctionState] = {}
43     self._active_calls: list[str] = []
44
45     def check(self, program: Program) -> SymbolTable:
46         for statement in program.statements:
47             if isinstance(statement, FunctionDef):
48                 self._register_function(statement)
49
50         for statement in program.statements:
51             if isinstance(statement, FunctionDef):
52                 continue
53             self._check_statement(statement, self.global_scope, None)
54
55         self._check_uncalled_functions()
56         return self.global_scope
57
58     def _register_function(self, node: FunctionDef) -> None:
59         if node.name in self._functions:
60             raise TypeCheckError(self._error_at(node, f"Function '{node
61                                     .name}' is already defined"))
62
63         placeholder_parameters = [(param_name, LanguageType.VOID) for
64                                   param_name in node.params]
65         try:
66             self.global_scope.define_function(node.name, LanguageType.
67                                               VOID, placeholder_parameters)
68         except SymbolTableError as error:
69             raise TypeCheckError(self._error_at(node, str(error))) from
70             error
71         self._functions[node.name] = _FunctionState(node=node)
72
73     def _check_statement(
74         self,
75         node: ASTNode,
76         scope: SymbolTable,
77         function_context: _FunctionContext | None,
78     ) -> None:
79         if isinstance(node, Assign):
80             value_type = self._infer_expr_type(node.value, scope)
81             self._assign_variable_type(scope, node, value_type)
82             return
83
84         if isinstance(node, Print):
85             self._infer_expr_type(node.expr, scope)
86             return
87
88         if isinstance(node, Return):
89             if function_context is None:
90                 raise TypeCheckError(self._error_at(node, "'return' is
91                                                         only valid inside a function"))
92             expr_type = self._infer_expr_type(node.expr, scope)
93             if function_context.return_type is None:

```

```

89         function_context.return_type = expr_type
90     elif function_context.return_type != expr_type:
91         raise TypeCheckError(
92             self._error_at(
93                 node,
94                 f"Function '{function_context.name}' returns
95                     both {function_context.return_type.value}
96                     and {expr_type.value}",
97             )
98         )
99     return
100
101     if isinstance(node, If):
102         condition_type = self._infer_expr_type(node.condition,
103             scope)
104         if condition_type != LanguageType.BOOLEAN:
105             raise TypeCheckError(self._error_at(node.condition, "If
106                 condition must have type Boolean"))
107         self._check_block(node.then_block, scope.child_scope(),
108             function_context)
109         if node.else_block is not None:
110             self._check_block(node.else_block, scope.child_scope(),
111                 function_context)
112         return
113
114     if isinstance(node, While):
115         condition_type = self._infer_expr_type(node.condition,
116             scope)
117         if condition_type != LanguageType.BOOLEAN:
118             raise TypeCheckError(self._error_at(node.condition, "
119                 While condition must have type Boolean"))
120         self._check_block(node.body, scope.child_scope(),
121             function_context)
122         return
123
124     if isinstance(node, Block):
125         self._check_block(node, scope.child_scope(),
126             function_context)
127         return
128
129     if isinstance(node, FunctionCall):
130         self._infer_function_call_type(node, scope)
131         return
132
133     raise TypeCheckError(self._error_at(node, f"Unsupported
134         statement node: {type(node).__name__}"))
135
136 def _check_block(
137     self,
138     block: Block,
139     scope: SymbolTable,
140     function_context: _FunctionContext | None,
141 ) -> None:

```

```

131         for statement in block.statements:
132             self._check_statement(statement, scope, function_context)
133
134     def _infer_expr_type(self, node: ASTNode, scope: SymbolTable) ->
LanguageType:
135         if isinstance(node, Literal):
136             return self._literal_type(node)
137
138         if isinstance(node, Identifier):
139             try:
140                 symbol = scope.lookup(node.name)
141             except SymbolTableError as error:
142                 raise TypeCheckError(self._error_at(node, str(error)))
from error
143             if not isinstance(symbol, VariableSymbol):
144                 raise TypeCheckError(self._error_at(node, f"'{node.name
}' is not a variable"))
145             return symbol.symbol_type
146
147         if isinstance(node, FunctionCall):
148             return self._infer_function_call_type(node, scope)
149
150         if isinstance(node, BinaryOp):
151             left_type = self._infer_expr_type(node.left, scope)
152             right_type = self._infer_expr_type(node.right, scope)
153             return self._infer_binary_type(node, left_type, right_type)
154
155         raise TypeCheckError(self._error_at(node, f"Unsupported
expression node: {type(node).__name__}"))
156
157     def _infer_function_call_type(self, node: FunctionCall, scope:
SymbolTable) -> LanguageType:
158         function_state = self._functions.get(node.name)
159         if function_state is None:
160             raise TypeCheckError(self._error_at(node, f"Undefined
function: {node.name}"))
161         if node.name in self._active_calls:
162             raise TypeCheckError(self._error_at(node, f"Recursive
function calls are not supported: {node.name}"))
163
164         argument_types = [self._infer_expr_type(argument, scope) for
argument in node.args]
165         function_symbol = self._lookup_function_symbol(node)
166
167         if len(argument_types) != len(function_symbol.parameters):
168             raise TypeCheckError(
169                 self._error_at(
170                     node,
171                     f"Function '{node.name}' expects {len(
function_symbol.parameters)} argument(s), got {
len(argument_types)}",
172                 )
173             )

```

```

174     declared_parameter_types = [param_type for _, param_type in
175                                function_symbol.parameters]
176     if all(param_type == LanguageType.VOID for param_type in
177            declared_parameter_types):
178         function_symbol.parameters = list(zip(function_state.node.
179                                                params, argument_types, strict=False))
180     else:
181         for index, ((_, expected_type), actual_type) in enumerate(
182             zip(function_symbol.parameters, argument_types, strict=
183                 False)):
184             if expected_type != actual_type:
185                 raise TypeCheckError(
186                     self._error_at(
187                         node.args[index],
188                         f"Argument {index + 1} of '{node.name}'
189                         must be {expected_type.value}, got {
190                             actual_type.value}",
191                     )
192                 )
193
194     if not function_state.body_checked:
195         self._check_function_body(function_state, function_symbol)
196
197     return function_symbol.symbol_type
198
199 def _check_function_body(self, function_state: _FunctionState,
200 function_symbol: FunctionSymbol) -> None:
201     function_context = _FunctionContext(name=function_state.node.
202                                         name)
203     function_scope = self.global_scope.child_scope()
204     for parameter_name, parameter_type in function_symbol.
205         parameters:
206         function_scope.define_variable(parameter_name,
207                                       parameter_type, initialized=True)
208
209     self._active_calls.append(function_state.node.name)
210     try:
211         self._check_block(function_state.node.body, function_scope,
212                           function_context)
213     finally:
214         self._active_calls.pop()
215
216     function_symbol.symbol_type = function_context.return_type or
217     LanguageType.VOID
218     function_state.body_checked = True
219
220 def _check_uncalled_functions(self) -> None:
221     """After all call-site checks, type-check any function whose
222     body was never visited."""
223     for name, state in self._functions.items():
224         if not state.body_checked:
225             try:

```

```

213         function_symbol = self.global_scope.lookup(name)
214     except SymbolTableError:
215         continue
216     if not isinstance(function_symbol, FunctionSymbol):
217         continue
218     inferred_params = self._infer_param_types_from_body(
219         state.node)
220     function_symbol.parameters = inferred_params
221     self._check_function_body(state, function_symbol)
222
223 def _infer_param_types_from_body(self, node: FunctionDef) -> list[
224     tuple[str, LanguageType]]:
225     """Infer parameter types by scanning the body for expression
226     contexts.
227
228     Any parameter used as an operand of an arithmetic or comparison
229     operator
230     is inferred as numeric (Float if the sibling operand is a float
231     literal,
232     Integer otherwise). Parameters whose usage gives no type hint
233     default to
234     Integer.
235     """
236     param_names = set(node.params)
237     inferred: dict[str, LanguageType] = {}
238
239 def scan_expr(n: ASTNode) -> None:
240     if isinstance(n, BinaryOp):
241         if n.op in {"+", "-", "*", "/", "=", "!="}:
242             for operand in (n.left, n.right):
243                 if isinstance(operand, Identifier) and operand.
244                     name in param_names:
245                     inferred.setdefault(operand.name,
246                         LanguageType.INTEGER)
247         elif n.op in {"+.", "-.", "*.", "/."}:
248             for operand in (n.left, n.right):
249                 if isinstance(operand, Identifier) and operand.
250                     name in param_names:
251                     inferred.setdefault(operand.name,
252                         LanguageType.FLOAT)
253         scan_expr(n.left)
254         scan_expr(n.right)
255     elif isinstance(n, FunctionCall):
256         for arg in n.args:
257             scan_expr(arg)
258
259 def scan_stmt(n: ASTNode) -> None:
260     if isinstance(n, Assign):
261         scan_expr(n.value)
262     elif isinstance(n, Print):
263         scan_expr(n.expr)
264     elif isinstance(n, Return):
265         scan_expr(n.expr)

```

```

256         elif isinstance(n, If):
257             scan_expr(n.condition)
258             scan_block(n.then_block)
259             if n.else_block:
260                 scan_block(n.else_block)
261         elif isinstance(n, While):
262             scan_expr(n.condition)
263             scan_block(n.body)
264         elif isinstance(n, Block):
265             scan_block(n)
266         elif isinstance(n, FunctionCall):
267             for arg in n.args:
268                 scan_expr(arg)
269
270     def scan_block(block: Block) -> None:
271         for stmt in block.statements:
272             scan_stmt(stmt)
273
274     scan_block(node.body)
275     return [(param, inferred.get(param, LanguageType.INTEGER)) for
276             param in node.params]
277
278     def _assign_variable_type(self, scope: SymbolTable, node: Assign,
279                               value_type: LanguageType) -> None:
280         try:
281             symbol = scope.lookup(node.name)
282         except SymbolTableError:
283             scope.define_variable(node.name, value_type, initialized=
284                                   True)
285             return
286
287         if not isinstance(symbol, VariableSymbol):
288             raise TypeCheckError(self._error_at(node, f"'{node.name}'
289                                                    is not a variable"))
290         if symbol.symbol_type != value_type:
291             raise TypeCheckError(
292                 self._error_at(
293                     node,
294                     f"Cannot assign {value_type.value} to variable '{
295                       node.name}' of type {symbol.symbol_type.value}",
296                 )
297             )
298         symbol.initialized = True
299
300     def _infer_binary_type(
301         self,
302         node: BinaryOp,
303         left_type: LanguageType,
304         right_type: LanguageType,
305     ) -> LanguageType:
306         # Integer operators: both operands must be Integer
307         if node.op in {"+", "-", "*"}:
308             if left_type != LanguageType.INTEGER or right_type !=

```

```

304         LanguageType.INTEGER:
305             raise TypeCheckError(
306                 self._error_at(node, f"Operator '{node.op}',
307                     requires two Integer operands")
308             )
309             return LanguageType.INTEGER
310
311     # Integer division: both operands must be Integer, result is
312     Integer
313     if node.op == "/":
314         if left_type != LanguageType.INTEGER or right_type !=
315             LanguageType.INTEGER:
316             raise TypeCheckError(
317                 self._error_at(node, f"Operator '{node.op}',
318                     requires two Integer operands")
319             )
320             return LanguageType.INTEGER
321
322     # Float operators: both operands must be Float
323     if node.op in {"+", "-", "*", "/"}:
324         if left_type != LanguageType.FLOAT or right_type !=
325             LanguageType.FLOAT:
326             raise TypeCheckError(
327                 self._error_at(node, f"Operator '{node.op}',
328                     requires two Float operands")
329             )
330             return LanguageType.FLOAT
331
332     if node.op in {"==", "!="}:
333         if not self._is_numeric(left_type) or not self._is_numeric(
334             right_type):
335             raise TypeCheckError(
336                 self._error_at(
337                     node,
338                     f"Operator '{node.op}' requires two arithmetic
339                     operands",
340                 )
341             )
342         if left_type != right_type:
343             raise TypeCheckError(
344                 self._error_at(
345                     node,
346                     f"Operator '{node.op}' requires operands of the
347                     same type, got {left_type.value} and {
348                         right_type.value}",
349                 )
350             )
351         return LanguageType.BOOLEAN
352
353     raise TypeCheckError(self._error_at(node, f"Unsupported
354         operator: {node.op}"))
355
356 def _lookup_function_symbol(self, node: FunctionCall) ->

```

```

FunctionSymbol:
345     try:
346         symbol = self.global_scope.lookup(node.name)
347     except SymbolTableError as error:
348         raise TypeCheckError(self._error_at(node, str(error))) from
            error
349     if not isinstance(symbol, FunctionSymbol):
350         raise TypeCheckError(self._error_at(node, f"'{node.name}',
            is not a function"))
351     return symbol
352
353     @staticmethod
354     def _is_numeric(language_type: LanguageType) -> bool:
355         return language_type in {LanguageType.INTEGER, LanguageType.
            FLOAT}
356
357     @staticmethod
358     def _literal_type(node: Literal) -> LanguageType:
359         if isinstance(node.value, bool):
360             return LanguageType.BOOLEAN
361         if isinstance(node.value, int):
362             return LanguageType.INTEGER
363         if isinstance(node.value, float):
364             return LanguageType.FLOAT
365         if isinstance(node.value, str):
366             return LanguageType.STRING
367         raise TypeCheckError(f"Unsupported literal value: {node.value!r}
            ")
368
369     @staticmethod
370     def _error_at(node: ASTNode, message: str) -> str:
371         return f"[line {node.line}, col {node.column}] TypeError: {
            message}"

```

Listing 8.8: components/type_checker.py — static type checker with inference

```

1 from __future__ import annotations
2
3 from dataclasses import dataclass
4 from typing import Callable
5
6 from components.ast_nodes import (
7     ASTNode,
8     Assign,
9     BinaryOp,
10    Block,
11    FunctionCall,
12    FunctionDef,
13    Identifier,
14    If,
15    Literal,
16    Print,
17    Program,

```



```

18     Return,
19     While,
20 )
21 from components.symbol_table import LanguageType
22
23
24 class InterpreterError(Exception):
25     pass
26
27
28 class _ReturnSignal(Exception):
29     def __init__(self, value: object) -> None:
30         self.value = value
31
32
33 @dataclass
34 class _FunctionRuntime:
35     node: FunctionDef
36
37
38 class _Environment:
39     def __init__(self, parent: _Environment | None = None) -> None:
40         self.parent = parent
41         self.values: dict[str, object] = {}
42
43     def define(self, name: str, value: object) -> None:
44         self.values[name] = value
45
46     def get(self, name: str) -> object:
47         if name in self.values:
48             return self.values[name]
49         if self.parent is not None:
50             return self.parent.get(name)
51         raise InterpreterError(f"Undefined variable: {name}")
52
53     def assign(self, name: str, value: object) -> None:
54         if name in self.values:
55             self.values[name] = value
56             return
57         if self.parent is not None and self.parent.contains(name):
58             self.parent.assign(name, value)
59             return
60         self.values[name] = value
61
62     def contains(self, name: str) -> bool:
63         if name in self.values:
64             return True
65         if self.parent is not None:
66             return self.parent.contains(name)
67         return False
68
69
70 class Interpreter:

```

```

71     def __init__(self, output_callback: Callable[[str], None] | None =
72         None) -> None:
73         self._functions: dict[str, _FunctionRuntime] = {}
74         self._output_callback = output_callback or print
75
76     def execute(self, program: Program) -> None:
77         environment = _Environment()
78         for statement in program.statements:
79             if isinstance(statement, FunctionDef):
80                 self._functions[statement.name] = _FunctionRuntime(node
81                     =statement)
82
83         for statement in program.statements:
84             if isinstance(statement, FunctionDef):
85                 continue
86             self._execute_statement(statement, environment)
87
88     def _execute_statement(self, node: ASTNode, environment:
89         _Environment) -> None:
90         if isinstance(node, Assign):
91             value = self._evaluate(node.value, environment)
92             environment.assign(node.name, value)
93             return
94
95         if isinstance(node, Print):
96             value = self._evaluate(node.expr, environment)
97             self._output_callback(f"{self._format_value(value)} : {self
98                 ._runtime_type(value).value}")
99             return
100
101         if isinstance(node, Return):
102             raise _ReturnSignal(self._evaluate(node.expr, environment))
103
104         if isinstance(node, If):
105             condition = self._evaluate(node.condition, environment)
106             if not isinstance(condition, bool):
107                 raise InterpreterError(self._error_at(node.condition, "
108                     If condition must evaluate to Boolean"))
109             branch = node.then_block if condition else node.else_block
110             if branch is not None:
111                 self._execute_block(branch, _Environment(parent=
112                     environment))
113             return
114
115         if isinstance(node, While):
116             while True:
117                 condition = self._evaluate(node.condition, environment)
118                 if not isinstance(condition, bool):
119                     raise InterpreterError(self._error_at(node.
120                         condition, "While condition must evaluate to
121                         Boolean"))
122                 if not condition:
123                     break

```

```

116         self._execute_block(node.body, _Environment(parent=
            environment))
117     return
118
119     if isinstance(node, Block):
120         self._execute_block(node, _Environment(parent=environment))
121         return
122
123     if isinstance(node, FunctionCall):
124         self._evaluate(node, environment)
125         return
126
127     raise InterpreterError(self._error_at(node, f"Unsupported
        statement node: {type(node).__name__}"))
128
129 def _execute_block(self, block: Block, environment: _Environment)
    -> None:
130     for statement in block.statements:
131         self._execute_statement(statement, environment)
132
133 def _evaluate(self, node: ASTNode, environment: _Environment) ->
    object:
134     if isinstance(node, Literal):
135         return node.value
136
137     if isinstance(node, Identifier):
138         return environment.get(node.name)
139
140     if isinstance(node, BinaryOp):
141         left_value = self._evaluate(node.left, environment)
142         right_value = self._evaluate(node.right, environment)
143         return self._evaluate_binary(node, left_value, right_value)
144
145     if isinstance(node, FunctionCall):
146         return self._call_function(node, environment)
147
148     raise InterpreterError(self._error_at(node, f"Unsupported
        expression node: {type(node).__name__}"))
149
150 def _call_function(self, node: FunctionCall, environment:
    _Environment) -> object:
151     function_runtime = self._functions.get(node.name)
152     if function_runtime is None:
153         raise InterpreterError(self._error_at(node, f"Undefined
            function: {node.name}"))
154
155     if len(node.args) != len(function_runtime.node.params):
156         raise InterpreterError(
157             self._error_at(
158                 node,
159                 f"Function '{node.name}' expects {len(
                    function_runtime.node.params)} argument(s), got
                    {len(node.args)}",

```

```

160         )
161     )
162
163     call_environment = _Environment(parent=environment)
164     argument_values = [self._evaluate(argument, environment) for
165         argument in node.args]
166     for parameter_name, argument_value in zip(function_runtime.node
167         .params, argument_values, strict=False):
168         call_environment.define(parameter_name, argument_value)
169
170     try:
171         self._execute_block(function_runtime.node.body,
172             call_environment)
173     except _ReturnSignal as signal:
174         return signal.value
175     return None
176
177 def _evaluate_binary(self, node: BinaryOp, left_value: object,
178     right_value: object) -> object:
179     if node.op == "+":
180         return left_value + right_value
181     if node.op == "-":
182         return left_value - right_value
183     if node.op == "*":
184         return left_value * right_value
185     if node.op == "/":
186         if int(right_value) == 0:
187             raise InterpreterError(self._error_at(node, "Division
188                 by zero"))
189         return int(left_value) // int(right_value)
190     if node.op == "+.":
191         return float(left_value) + float(right_value)
192     if node.op == "-.":
193         return float(left_value) - float(right_value)
194     if node.op == ".*":
195         return float(left_value) * float(right_value)
196     if node.op == "/.":
197         if float(right_value) == 0.0:
198             raise InterpreterError(self._error_at(node, "Division
199                 by zero"))
200         return float(left_value) / float(right_value)
201     if node.op == "==":
202         return left_value == right_value
203     if node.op == "!=":
204         return left_value != right_value
205     raise InterpreterError(self._error_at(node, f"Unsupported
206         operator: {node.op}"))
207
208 @staticmethod
209 def _format_value(value: object) -> str:
210     if isinstance(value, str):
211         return f'"{value}"'
212     if isinstance(value, bool):

```

```

206         return "true" if value else "false"
207         return str(value)
208
209     @staticmethod
210     def _runtime_type(value: object) -> LanguageType:
211         if isinstance(value, bool):
212             return LanguageType.BOOLEAN
213         if isinstance(value, int):
214             return LanguageType.INTEGER
215         if isinstance(value, float):
216             return LanguageType.FLOAT
217         if isinstance(value, str):
218             return LanguageType.STRING
219         return LanguageType.VOID
220
221     @staticmethod
222     def _error_at(node: ASTNode, message: str) -> str:
223         return f"[line {node.line}, col {node.column}] RuntimeError: {
            message}"

```

Listing 8.9: components/interpreter.py — tree-walking interpreter

```

1  from __future__ import annotations
2
3  from dataclasses import dataclass
4
5  from components.ast_printer import ASTPrinter
6  from components.interpreter import Interpreter
7  from components.lexica import Lexer
8  from components.parser import Parser
9  from components.symbol_table import SymbolTable
10 from components.tokens import Token
11 from components.type_checker import TypeChecker
12
13
14 @dataclass
15 class PipelineResult:
16     tokens: list[Token]
17     ast_text: str
18     checked_scope: SymbolTable
19     outputs: list[str]
20
21
22 def run_pipeline(source_code: str) -> PipelineResult:
23     tokens = Lexer(source_code).tokenize()
24     tree = Parser(tokens).parse()
25     ast_text = ASTPrinter().print(tree)
26     checked_scope = TypeChecker().check(tree)
27
28     outputs: list[str] = []
29     Interpreter(output_callback=outputs.append).execute(tree)
30
31     return PipelineResult(

```

```

32         tokens=tokens,
33         ast_text=ast_text,
34         checked_scope=checked_scope,
35         outputs=outputs,
36     )
37
38
39 def format_tokens(tokens: list[Token]) -> str:
40     lines = []
41     for token in tokens:
42         lines.append(
43             f"    {token.token_type.name:<14} {token.lexeme!r:<12} line={
44                 token.line} col={token.column}"
45         )
46     return "\n".join(lines)
47
48 def format_stage_output(result: PipelineResult) -> str:
49     sections = [
50         "=" * 60,
51         "    STAGE 1 -- TOKENS",
52         "=" * 60,
53         format_tokens(result.tokens),
54         "",
55         "=" * 60,
56         "    STAGE 2 -- AST (parser output)",
57         "=" * 60,
58         result.ast_text,
59         "",
60         "=" * 60,
61         "    STAGE 3 -- TYPE CHECK",
62         "=" * 60,
63         result.checked_scope.format_table(),
64         "",
65         "=" * 60,
66         "    STAGE 4 -- EXECUTION",
67         "=" * 60,
68         "\n".join(result.outputs) if result.outputs else "(no output)",
69     ]
70     return "\n".join(sections)

```

Listing 8.10: components/pipeline.py — shared pipeline (CLI, UI, and tests)

A.3 Test Suite

```

1 from __future__ import annotations
2
3 import unittest
4
5 from components.lexica import Lexer, LexerError
6 from components.parser import Parser, ParseError
7 from components.pipeline import run_pipeline

```

```

8 from components.symbol_table import LanguageType, SymbolTable,
   SymbolTableError
9 from components.tokens import TokenType
10 from components.type_checker import TypeCheckError
11
12
13 #
   -----
14 # Lexer
15 #
   -----
16
17 class LexerTests(unittest.TestCase):
18     """Tests for components/lexica.py -- lexical analysis and
        tokenisation."""
19
20     def test_integer_literal_token(self) -> None:
21         tokens = Lexer("42").tokenize()
22         self.assertEqual(tokens[0].token_type, TokenType.INT_LITERAL)
23         self.assertEqual(tokens[0].lexeme, "42")
24
25     def test_float_literal_token(self) -> None:
26         tokens = Lexer("3.14").tokenize()
27         self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
28         self.assertEqual(tokens[0].lexeme, "3.14")
29
30     def test_bool_literals_produce_bool_literal_tokens(self) -> None:
31         tokens = Lexer("true false").tokenize()
32         self.assertEqual(tokens[0].token_type, TokenType.BOOL_LITERAL)
33         self.assertEqual(tokens[0].lexeme, "true")
34         self.assertEqual(tokens[1].token_type, TokenType.BOOL_LITERAL)
35         self.assertEqual(tokens[1].lexeme, "false")
36
37     def test_keywords_produce_correct_token_types(self) -> None:
38         tokens = Lexer("if else while def return print").tokenize()
39         expected = [
40             TokenType.IF, TokenType.ELSE, TokenType.WHILE,
41             TokenType.DEF, TokenType.RETURN, TokenType.PRINT,
42         ]
43         actual = [t.token_type for t in tokens if t.token_type !=
44                   TokenType.EOF]
45         self.assertEqual(actual, expected)
46
47     def test_identifier_not_confused_with_keyword(self) -> None:
48         tokens = Lexer("iffy whileloop").tokenize()
49         self.assertEqual(tokens[0].token_type, TokenType.IDENTIFIER)
50         self.assertEqual(tokens[1].token_type, TokenType.IDENTIFIER)
51
52     def test_two_character_operators(self) -> None:
53         tokens = Lexer("== !=").tokenize()
54         self.assertEqual(tokens[0].token_type, TokenType.EQUAL_EQUAL)

```

```

54         self.assertEqual(tokens[1].token_type, TokenType.BANG_EQUAL)
55
56     def test_float_dot_operators_tokenised(self) -> None:
57         tokens = Lexer("+. -. *. /.").tokenize()
58         expected = [TokenType.PLUS_DOT, TokenType.MINUS_DOT, TokenType.
59                     STAR_DOT, TokenType.SLASH_DOT]
60         actual = [t.token_type for t in tokens if t.token_type !=
61                  TokenType.EOF]
62         self.assertEqual(actual, expected)
63
64     def test_float_literal_not_confused_with_dot_operator(self) -> None
65     :
66         # "3.14" must produce FLOAT_LITERAL, not INT_LITERAL followed
67         # by something
68         tokens = Lexer("3.14").tokenize()
69         self.assertEqual(tokens[0].token_type, TokenType.FLOAT_LITERAL)
70         self.assertEqual(tokens[0].lexeme, "3.14")
71
72     def test_source_position_tracked_across_lines(self) -> None:
73         tokens = Lexer("x\ny").tokenize()
74         self.assertEqual(tokens[0].line, 1)
75         self.assertEqual(tokens[1].line, 2)
76
77     def test_unexpected_character_raises_lexer_error(self) -> None:
78         with self.assertRaises(LexerError):
79             Lexer("@").tokenize()
80
81     def test_lone_exclamation_raises_lexer_error(self) -> None:
82         with self.assertRaises(LexerError):
83             Lexer("!5").tokenize()
84
85     def test_unterminated_string_raises_lexer_error(self) -> None:
86         with self.assertRaises(LexerError):
87             Lexer('"hello').tokenize()
88
89     #
90     -----
91
92     # Symbol table
93     #
94     -----
95
96     class SymbolTableTests(unittest.TestCase):
97         """Tests for components/symbol_table.py -- scoped symbol table and
98         semantic types."""
99
100         def test_define_variable_and_lookup(self) -> None:
101             table = SymbolTable()
102             table.define_variable("x", LanguageType.INTEGER, initialized=
103                                 True)
104             symbol = table.lookup("x")

```



```

97         self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
98
99     def test_duplicate_variable_definition_raises_error(self) -> None:
100         table = SymbolTable()
101         table.define_variable("x", LanguageType.INTEGER)
102         with self.assertRaises(SymbolTableError):
103             table.define_variable("x", LanguageType.FLOAT)
104
105     def test_duplicate_function_definition_raises_error(self) -> None:
106         table = SymbolTable()
107         table.define_function("f", LanguageType.INTEGER, [])
108         with self.assertRaises(SymbolTableError):
109             table.define_function("f", LanguageType.FLOAT, [])
110
111     def test_lookup_in_parent_scope(self) -> None:
112         parent = SymbolTable()
113         parent.define_variable("x", LanguageType.INTEGER, initialized=
            True)
114         child = parent.child_scope()
115         symbol = child.lookup("x")
116         self.assertEqual(symbol.symbol_type, LanguageType.INTEGER)
117
118     def test_undefined_symbol_raises_error(self) -> None:
119         table = SymbolTable()
120         with self.assertRaises(SymbolTableError):
121             table.lookup("z")
122
123     def test_format_table_contains_variable_row(self) -> None:
124         table = SymbolTable()
125         table.define_variable("counter", LanguageType.INTEGER,
            initialized=True)
126         output = table.format_table()
127         self.assertIn("counter", output)
128         self.assertIn("variable", output)
129         self.assertIn("Integer", output)
130
131     def test_format_table_contains_function_row(self) -> None:
132         table = SymbolTable()
133         table.define_function("add", LanguageType.INTEGER, [("a",
            LanguageType.INTEGER)])
134         output = table.format_table()
135         self.assertIn("add", output)
136         self.assertIn("function", output)
137
138
139 #
140 # Parser
141 #

```

```

143 class ParserTests(unittest.TestCase):
144     """Tests for components/parser.py -- recursive-descent parsing."""
145
146     def _parse(self, source: str):
147         return Parser(Lexer(source).tokenize()).parse()
148
149     def test_missing_semicolon_raises_parse_error(self) -> None:
150         with self.assertRaises(ParseError):
151             self._parse("x = 5")
152
153     def test_unclosed_brace_raises_parse_error(self) -> None:
154         with self.assertRaises(ParseError):
155             self._parse("if (true) { x = 1;}")
156
157     def test_operator_precedence_mul_before_add(self) -> None:
158         # 2 + 3 * 4 must be 14, not 20
159         result = run_pipeline("x = 2 + 3 * 4; print(x);")
160         self.assertEqual(result.outputs, ["14 : Integer"])
161
162     def test_operator_left_associativity(self) -> None:
163         # 10 - 3 - 2 must be 5 (left-assoc), not 9
164         result = run_pipeline("x = 10 - 3 - 2; print(x);")
165         self.assertEqual(result.outputs, ["5 : Integer"])
166
167     def test_parentheses_override_precedence(self) -> None:
168         # (2 + 3) * 4 must be 20
169         result = run_pipeline("x = (2 + 3) * 4; print(x);")
170         self.assertEqual(result.outputs, ["20 : Integer"])
171
172
173 #
174 # -----
175 #
176
177 class TypeCheckerTests(unittest.TestCase):
178     """Tests for components/type_checker.py -- static type inference and
179     checking."""
180
181     def test_integer_arithmetic_stays_integer(self) -> None:
182         result = run_pipeline("x = 3 + 4; print(x);")
183         self.assertEqual(result.outputs, ["7 : Integer"])
184
185     def test_division_always_produces_integer(self) -> None:
186         # integer / integer -> Integer (no implicit float conversion)
187         result = run_pipeline("x = 10 / 2; print(x);")
188         self.assertEqual(result.outputs, ["5 : Integer"])
189
190     def test_float_dot_operators_produce_float(self) -> None:
191         result = run_pipeline("x = 1.0 +. 2.5; print(x);")

```

```

191         self.assertEqual(result.outputs, ["3.5 : Float"])
192
193     def test_mixed_int_float_addition_raises_type_error(self) -> None:
194         # Integer + Float is no longer valid; must use +.
195         with self.assertRaises(TypeError):
196             run_pipeline("x = 1 + 2.5; print(x);")
197
198     def test_string_variable_type_inferred(self) -> None:
199         result = run_pipeline('x = "hello"; print(x);')
200         self.assertEqual(result.outputs, ['"hello" : String'])
201
202     def test_boolean_literal_type_inferred(self) -> None:
203         result = run_pipeline("x = true; print(x);")
204         self.assertEqual(result.outputs, ["true : Boolean"])
205
206     def test_boolean_equality_raises_type_error(self) -> None:
207         # Boolean == Boolean is not allowed; == only accepts arithmetic
208         operands
209         with self.assertRaises(TypeError):
210             run_pipeline('x = true; if (x == false) { print("then"); }'
211                           'else { print("else"); }')
212
213     def test_mixed_int_float_equality_raises_type_error(self) -> None:
214         # == and != must not accept mixed Integer/Float (no overloading
215         )
216         with self.assertRaises(TypeError):
217             run_pipeline("x = 5; y = 3.14; if (x == y) { print(x); }")
218
219     def test_mixed_int_float_inequality_raises_type_error(self) -> None:
220         :
221         with self.assertRaises(TypeError):
222             run_pipeline("x = 5; y = 3.14; if (x != y) { print(x); }")
223
224     def test_arithmetic_on_string_raises_type_error(self) -> None:
225         with self.assertRaises(TypeError):
226             run_pipeline('x = "a" + 1;')
227
228     def test_if_non_boolean_condition_raises_type_error(self) -> None:
229         with self.assertRaises(TypeError):
230             run_pipeline("if (1) { x = 2; }")
231
232     def test_while_non_boolean_condition_raises_type_error(self) ->
233     None:
234         with self.assertRaises(TypeError):
235             run_pipeline("x = 0; while (x) { x = x + 1; }")
236
237     def test_assignment_type_mismatch_raises_type_error(self) -> None:
238         with self.assertRaises(TypeError):
239             run_pipeline('x = 5; x = "hello";')
240
241     def test_undefined_variable_raises_type_error(self) -> None:
242         with self.assertRaises(TypeError):
243             run_pipeline("print(z);")

```

```

239
240 def test_function_wrong_arg_count_raises_type_error(self) -> None:
241     with self.assertRaises(TypeError):
242         run_pipeline("def f(a) { return a; } f(1, 2);")
243
244 def test_function_arg_type_mismatch_raises_type_error(self) -> None
245 :
246     # first call infers a: Integer; second call with String must
247     fail
248     with self.assertRaises(TypeError):
249         run_pipeline('def f(a) { return a; } f(1); f("hello");')
250
251 def test_uncalled_function_body_type_error_is_caught(self) -> None:
252     # type error inside body must be detected even if function is
253     never called
254     with self.assertRaises(TypeError):
255         run_pipeline('def bad() { y = "hello" + 1; }')
256
257 def test_valid_uncalled_function_does_not_raise(self) -> None:
258     # a well-typed but uncalled function should not raise
259     result = run_pipeline("def add(a, b) { return a + b; }")
260     self.assertEqual(result.outputs, [])
261
262 #
263 -----
264
265 # Integration (full pipeline)
266 #
267 -----
268
269
270 class IntegrationTests(unittest.TestCase):
271     """End-to-end tests through all four pipeline stages."""
272
273     # --- Unary minus ---
274
275     def test_unary_minus_integer(self) -> None:
276         result = run_pipeline("x = -5; print(x);")
277         self.assertEqual(result.outputs, ["-5 : Integer"])
278
279     def test_unary_minus_float(self) -> None:
280         result = run_pipeline("y = -. 3.14; print(y);")
281         self.assertEqual(result.outputs, ["-3.14 : Float"])
282
283     def test_unary_minus_variable(self) -> None:
284         result = run_pipeline("x = 10; y = -x; print(y);")
285         self.assertEqual(result.outputs, ["-10 : Integer"])
286
287     def test_unary_minus_inside_expression(self) -> None:
288         result = run_pipeline("x = 3 + -2; print(x);")
289         self.assertEqual(result.outputs, ["1 : Integer"])

```

```

285 # --- Control flow ---
286
287 def test_if_then_branch_executes(self) -> None:
288     result = run_pipeline(
289         'x = 5; if (x == 5) { print("yes"); } else { print("no"); }
290     )
291     self.assertEqual(result.outputs, ['"yes" : String'])
292
293 def test_if_else_branch_executes_when_condition_false(self) -> None:
294     result = run_pipeline(
295         'x = 0; if (x == 5) { print("yes"); } else { print("no"); }
296     )
297     self.assertEqual(result.outputs, ['"no" : String'])
298
299 def test_while_loop_executes_repeatedly(self) -> None:
300     result = run_pipeline("x = 3; while (x != 0) { print(x); x = x
301         - 1; }")
302     self.assertEqual(result.outputs, ["3 : Integer", "2 : Integer",
303         "1 : Integer"])
304
305 # --- Functions ---
306
307 def test_function_integer_return(self) -> None:
308     result = run_pipeline("def square(n) { return n * n; } print(
309         square(7));")
310     self.assertEqual(result.outputs, ["49 : Integer"])
311
312 def test_function_type_inference(self) -> None:
313     result = run_pipeline(
314         "def add(a, b) { return a +. b; } result = add(2.0, 3.5);
315         print(result);"
316     )
317     self.assertEqual(result.outputs, ["5.5 : Float"])
318     self.assertIn("result      variable   Float", result.
319         checked_scope.format_table())
320
321 def test_function_value_parameter_passing(self) -> None:
322     # Modifying a parameter inside a function must not affect the
323     # caller's variable
324     result = run_pipeline(
325         "def inc(a) { a = a + 1; return a; } x = 10; y = inc(x);
326         print(x); print(y);"
327     )
328     self.assertEqual(result.outputs, ["10 : Integer", "11 : Integer
329         "])
330
331 def test_nested_function_calls(self) -> None:
332     result = run_pipeline(
333         "def add(a, b) { return a + b; } print(add(add(1, 2), 3));"
334     )

```

```

327         self.assertEqual(result.outputs, ["6 : Integer"])
328
329     # --- print() with all four types ---
330
331     def test_print_integer(self) -> None:
332         result = run_pipeline("print(42);")
333         self.assertEqual(result.outputs, ["42 : Integer"])
334
335     def test_print_float(self) -> None:
336         result = run_pipeline("print(3.14);")
337         self.assertEqual(result.outputs, ["3.14 : Float"])
338
339     def test_print_boolean(self) -> None:
340         result = run_pipeline("print(true);")
341         self.assertEqual(result.outputs, ["true : Boolean"])
342
343     def test_print_string(self) -> None:
344         result = run_pipeline('print("plc");')
345         self.assertEqual(result.outputs, ['"plc" : String'])
346
347     # --- Reassignment ---
348
349     def test_variable_can_be_reassigned_same_type(self) -> None:
350         result = run_pipeline("x = 1; x = 2; x = 3; print(x);")
351         self.assertEqual(result.outputs, ["3 : Integer"])
352
353     def test_scope_isolation_if_block(self) -> None:
354         # Variable defined inside an if-block must not be visible
355         # outside it
356         with self.assertRaises(TypeError):
357             run_pipeline(
358                 "x = 1; if (x == 1) { inner = 99; } print(inner);"
359             )
360
361 if __name__ == "__main__":
362     unittest.main()

```

Listing 8.11: tests/test_pipeline.py — automated regression suite (59 tests)